



A Quantum-Enhanced Deep Learning Framework for Precise Leaf Disease Detection in Tomato and Cotton Crops

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Abstract

Leaf diseases significantly threaten agricultural yields, making their prompt and precise detection crucial for effective management and crop safeguarding. This work aims to create a strong classification framework for detecting leaf diseases in crops by utilizing advanced deep learning methods. The work uses the Dual-Tree Complex Wavelet Preprocessing Framework (DTCWPF) to standardize and reduce noise in input images by maintaining key features for accurate analysis. Feature extraction has been conducted using a blend of EfficientNet-B7 and Inception-ResNet v2 models for capturing a wide range of image attributes such as color, texture, shape, and structural details from both RGB and grayscale images. The Archimedes Optimization Algorithm (AOA) is then applied to select the most pertinent features and to enhance the classification performance. The framework integrates quantum computing principles by employing an Attention-Based Quantum Convolutional Neural Network (ABQCNN), coupled with Adaptive Silver Fox Algorithm (ASFA) for classification. This combination enhances the ability of the model to effectively prioritize salient data features and adaptively optimize feature selection, thereby improving the overall classification performance. To ensure the interpretability of the model predictions, GradCAM and GradCAM++ techniques are applied to generate class-discriminative visual explanations and for highlighting the critical regions within the leaf images those most significantly influence the classification outcomes. These visualization strategies provide valuable insights into the internal decision-making process of the model and subsequently, substantiate its robustness and reliability. The proposed framework has been evaluated on two tomato leaf datasets and two cotton leaf datasets and outstanding performances have been depicted. For the tomato leaf datasets, the model has achieved accuracies of 99.51% and 99.70%, respectively by demonstrating its robustness in classifying various tomato leaf diseases. Similarly, on the cotton leaf datasets, the model has achieved accuracies of 99.92% and 99.60%, respectively by demonstrating its effectiveness in distinguishing different cotton leaf conditions and highlighted the versatility of the framework as well as its potential application in real-world agricultural disease detection scenarios, thereby offering significant advancements in plant pathology.

Keywords Preprocessing · CNN · ASFA · AOA · EfficientNet-B7 · Inception-ResNet v2

1 Introduction

The agricultural domain has increasingly integrated advanced technologies to pursue sustainable and efficient farming (Yao et al. 2024a, b; Li et al. 2024a, b). The major challenge faced globally is the early and accurate identification of leaf-based crop diseases, which, if undetected, can lead to serious yield losses and economic damage (Zhang et al. 2024; Ritharson et al. 2024). Conventional diagnostic methods that depend on manual inspection are not only time-consuming but also error-prone, prompting a shift toward Artificial Intelligence (AI)-driven solutions, particularly deep and machine learning (Li et al. 2024a, b). Leaf diseases typically appear as spots, rust, blights, and molds, which

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are manifestations of infections caused by various pathogens such as fungi, bacteria, or viruses (Trinh et al. 2024; Upadhyay et al. 2025). These pathogens impair plant health by disrupting photosynthesis, often resulting in wilting, discoloration, lesions, and other deformities (Lv and Wen-Hao 2024; Fu et al. 2024; Diker et al. 2024). An early diagnosis is essential for timely intervention and containment.

Image-based disease detection generally follows a multi-step pipeline: high-quality image acquisition, preprocessing to enhance clarity, and segmentation to separate diseased and healthy tissues (Elfatimi et al. 2024). Deep learning models are then employed to extract features and perform classification, thereby offering reliable results across large datasets (Bonkra et al. 2024). However, its adaptability remains a critical issue, and variations in climate, soil, cultivation practices, and imaging conditions such as lighting or background clutter often reduce the generalizability of the model (Khanna et al. 2024). Another key limitation is the scarcity of diverse, standardized datasets encompassing multi-crop scenarios, which restrict the performance in field applications.

To overcome these shortcomings, this work has introduced a novel hybrid framework that combines classical deep learning with quantum-enhanced architectures. The model employs an Attention-Based Quantum Convolutional Neural Network (ABQCNN) which integrates quantum-inspired convolution and pooling operations with classical layers. These quantum elements simulate entanglement and superposition, enabling the proposed model to extract complex feature relationships with higher precision and efficiency than the standard CNNs.

The proposed system has been designed to be robust across datasets, lightweight enough for deployment, and effective without demanding high computational resources or expert tuning. Embedding quantum attention mechanisms and leveraging optimization strategies improve the focus on disease-affected regions. This approach bridges theoretical advancements with field-level practicality and offers real-time, scalable diagnostics that align with modern agricultural needs (Jha et al. 2024; Lanjewar et al. 2024). Overall, this study introduces a distinct methodological shift by combining quantum computing concepts with deep learning to enhance the detection performance in diverse agricultural environments and it offers improved accuracy, adaptability, and deployment feasibility by providing a promising direction for future research in intelligent crop-monitoring systems (Usha Ruby et al. 2024). The primary objective of this research is to develop an efficient and accurate framework for leaf disease identification by integrating advanced techniques with image preprocessing, feature extraction, and feature selection. Specifically, this study is aimed to:

- Enhance image preprocessing using a Dual-Tree Complex Wavelet Preprocessing Framework (DTCWPF) for standardizing and to denoise the inputs.
- Extract comprehensive and relevant features from the RGB and grayscale images by using EfficientNet-B7 and Inception-ResNet v2.
- Optimize feature selection using the Archimedes Optimization Algorithm (AOA) and to improve the classification accuracy of the machine learning models.
- Incorporate quantum computing principles via an Attention-Based Quantum Convolutional Neural Network (ABQCNN) and to further enhance the model's classification accuracy.
- Leverage the Adaptive Silver Fox Algorithm (ASFA) in conjunction with other methodologies to further optimize the classification performance for ensuring high accuracy and reliability across various datasets.
- Apply GradCAM and GradCAM++ techniques to generate visual explanations of the model's focus areas and to enhance the interpretability and transparency of the classification process by identifying the key regions in the leaf images that contribute to the model's predictions.

The rest of this paper is organized as follows. Section 2 delves into the literature for emphasizing both the advancements and the challenges in identifying plant leaf diseases, particularly through the use of quantum-inspired approaches and deep-learning methods. Section 3 details the methodology including preprocessing, feature extraction, selection, classification, and optimization strategies within the proposed framework. Section 4 outlines the experimental setup as well as the results and the performance metrics such as classification accuracy and computational efficiency are evaluated across various datasets. Finally, Sect. 5 describes the findings and proposes future directions, including the incorporation of quantum-enhanced models for scalable and adaptable agricultural solutions.

2 Related Works

Preprocessing is a foundational step in standardizing input data, where images are resized (e.g., 224×224 or 256×256 pixels) and enhanced through Gaussian or median filtering to suppress noise and improve clarity (Rehman et al. 2025; Omer et al. 2024). Data augmentation strategies, such as rotation, flipping, shifting, and noise addition, are widely employed to expand the dataset diversity and combat overfitting. For complex backgrounds, segmentation is done using K-means clustering and optimized via Grey Wolf Optimization (GWO) which helps to isolate leaf regions and

effectively manage overlapping pixel boundaries (Deari and Ulukaya 2024; Kusuma 2024).

Feature extraction enables the model to learn relevant visual patterns. CNNs remain central to this task because of their ability to identify spatial hierarchies (Alam et al. 2024; Yao et al. 2024a, b), whereas advanced models such as 3D-CNN and Capsule Networks (CapsNet) capture both the spatial and the temporal cues (Abouelmagd et al. 2024). Modified Xception-based multilevel feature fusion (MX-MLF2) improves pattern learning by combining intermediate-level features (Thangaraj, Rajasekaran et al. 2024). Traditional descriptors such as GLCM and LBP, when integrated with deep networks, offer richer texture and shape information (Kaur and Devendran 2024). Quantum CNNs (QCNNs) have recently attracted interest, due to their capability to handle intricate representations with fewer parameters and improved generalization.

To optimize the performance, feature selection techniques such as SVM-RFE and Adaptive Rain Optimization (ARO) are employed to reduce redundancy and highlight informative attributes (Bhagat and Kumar 2024; Dubey et al. 2024a, b). Quantum-inspired selection techniques have shown improved stability and efficiency compared to the classical methods, particularly for high-dimensional data. Hybrid and ensemble models have gained prominence in classification. For instance, combining 3D-CNNs with LSTM networks allows spatiotemporal dependencies to be captured (Ashwini and Sellam 2024; Dubey et al. 2024a, b). Ensemble Stacked Deep Neural Networks (ESDNNs) aggregate outputs from multiple models to improve classification robustness (Gautam et al. 2024). Transfer learning with pre-trained networks, such as VGG16, InceptionV3, ResNet50, and DenseNet121, has also significantly boosted accuracy across various crop datasets (Acharya and Ravi 2024).

Optimization algorithms further refine the learning and stability. Techniques such as Improved Backtracking Search (IBS) and Hunter-Prey Optimization (Hunt-PO) are used to avoid issues such as gradient vanishing and premature convergence. Recently, quantum optimization techniques that leverage quantum-enhanced algorithms have been proposed for hyperparameter tuning and global optimization (Stephen, Ancy et al. 2024; Singla et al. 2024). For instance, the Spotted Hyena Archimedes Optimizer (SHAO) has been shown to significantly enhance classification accuracy, making it suitable for real-time agricultural applications (Tripathy and Mandala 2024). Table 1 shows the comparison of the existing techniques. While these developments have improved the accuracy and deployment feasibility, quantum-based models remain a growing research area, and they offer potential advantages in terms of scalability, speed, and adaptability. This study is built on prior work by combining quantum-enhanced techniques with classical deep learning

to close the existing gaps in robustness and generalization in plant leaf disease detection systems.

3 Methodology

This research has utilized sophisticated methods, including image preprocessing, feature extraction, and feature selection, to enhance the precision of leaf disease identification, as depicted in Fig. 1. The novelty of this study lies in the strategic combination and adaptation of cutting-edge techniques to address specific challenges in agricultural disease detection rather than introducing new elements within the individual components. The combination of EfficientNet-B7 and Inception-ResNet v2 enables simultaneous analysis of RGB and grayscale images and facilitates enhanced feature extraction that supports the classification of complex datasets. This comprehensive method has not been widely explored in the realm of agricultural disease classification and hence, the novel elements of this framework are emphasized. The inclusion of quantum computing principles through the Attention-Based Quantum Convolutional Neural Network (ABQCNN) further distinguishes this study. Quantum-inspired mechanisms, such as quantum convolution and pooling, have been implemented in simulation environments and the capability to process intricate feature interactions with reduced computational requirements has been demonstrated. While the practical application on quantum hardware is still limited by current technological constraints, these simulations offer insights into the potential improvements in efficiency and accuracy that quantum-inspired methods could bring. Future efforts will focus on expanding the experimental validation and providing detailed comparisons to strengthen these claims. This study aims to advance the existing methodologies by combining well-established techniques in a novel manner and introducing emerging computational paradigms to the field.

3.1 Dual-Tree Complex Wavelet Preprocessing Framework (DTCWPF)

The Dual-Tree Complex Wavelet Transform (DTCWT) extends the capabilities of the traditional wavelet frameworks by utilizing two parallel filter banks to generate complex coefficients that capture both amplitude and phase information. This dual structure enables near-shift-invariance and directional sensitivity for allowing the transform to respond selectively to edges and textures at multiple orientations, including $\pm 15^\circ$, $\pm 45^\circ$, and $\pm 75^\circ$. Such properties are especially advantageous in plant disease scenarios, where irregularities, such as mottling, edge roughness, and vein deformation, often emerge in specific directions. Through multilevel decomposition, the DTCWT preserves

Table 1 Summary of the existing Literature Survey in leaf disease detection

References No	Authors	Algorithms	Key Advantages	Limitations
[2]	(Li, et al., 2024)	A lightweight hybrid CNN-Transformer network (U-RSwinT)	Reduced training time by 51.1% compared to LeafGAN	Potential misclassifications in diseases with similar lesions
[5]	(Li, et al., 2024)	Transformer-based model incorporating CBAM and EQL loss functions	Achieved high detection accuracy by effectively addressing class imbalance	high model complexity could affect real-time applicability.
[10]	(Diker, et al., 2024)	Detecting peacock eye disease in olive leaves using CNN models and transfer learning (ResNet101 and MobileNet)	Achieved high classification accuracy of 98.63% using a combination of CNN- and classifiers like Random Forest and SVM	The study was limited to only two classes (infected and healthy), and the dataset size was relatively small (954 samples)
[23]	(Abouelmagd, et al., 2024)	Optimized Capsule Neural Network (CapsNet) with Adam optimizer	Achieved high accuracy of 96.39% in classifying ten different tomato leaf diseases	Limited to a specific dataset and did not utilize UAV images for testing
[33]	(Singla, et al., 2024)	A deep CNN algorithm was trained using the Plant Village dataset to detect the diseases in plant leaves	Attained an accuracy rate of 99.39% in binary classification	The suggested method was tested on laboratory-condition images
[9]	(Fu, et al., 2024)	A lightweight CNN algorithm, GGM-VGG16, incorporated with Ghost modules and global average pooling for recognizing pepper leaf diseases in images	Attained perfect accuracy on images featuring human palm backgrounds and maintained a lightweight architecture with a size of just 11.5 MB, highlighting its suitability for deployment in mobile-based applications	Accuracy dropped to 87.38% on images with non-palm backgrounds, indicating a need for improvement in generalization to diverse backgrounds
[8]	(Lv, and Wen-Hao, 2024)	YOLOV5-CBAM-C3TR with transformer encoder for apple leaf disease	Accomplished a mean Average Precision (mAP) of 73.4% at an IoU threshold of 0.5 and 40.9% across the IoU range of 0.5 to 0.95, along with an average accuracy of 92.4% in real-world testing scenarios	Increased the model complexity and parameter count, which affected the deployment in resource-constrained environments
[30]	(Gautam, et al., 2023)	An ensemble stacked deep neural network algorithm with logistic regression for mango leaf disease	Reached a classification accuracy of 98.57%, demonstrating the model's high performance	The suggested algorithm considered biotic diseases and it is needed to be expanded for abiotic factors
[14]	(Jha, and Dembla, 2023)	A transfer learning-based stacking ensemble algorithm using MobileNet, Inception, and ResNet for potato leaf disease	Achieved a classification accuracy of 98.86%	Limited to three specific pre-trained models; further improvement is needed by training on a larger dataset
[19]	(Deari, and Ulukaya, 2023)	A hybrid Inception v3 and YOLOv5x for rice leaf disease	Obtained excellent results with an accuracy of 96.67% and an F1 score of 98.24%	Increased computational complexity, due to the combination of classification and detection networks

both the coarse and the fine details without introducing significant distortion or aliasing. Its inverse mechanism recombines processed sub-bands using complementary filters, ensuring that the important structural cues remain intact. By reducing the redundant frequency content and enhancing discriminative spatial features, this approach offers a preprocessing pathway that is not only robust, but also

computationally efficient for downstream analysis. Image preprocessing plays a critical role in preparing visual data for analysis, particularly in the context of leaf disease identification. It transforms raw image data into a standardized format by addressing variations in the image quality, lighting, and dimensions. This guarantees that the input data are consistent and standardized and it is crucial for the precise

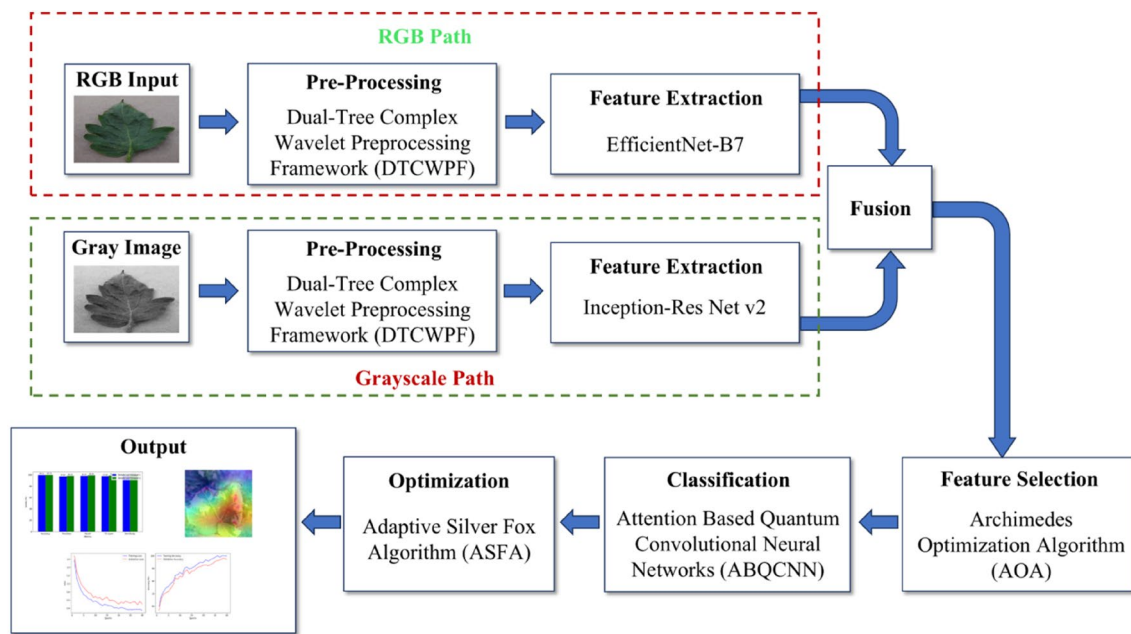


Fig. 1 Proposed block diagram

and effective functioning of deep learning models. The pre-processing techniques employed in this study are outlined below.

To ensure uniformity in input size and compatibility with neural network designs, images are adjusted to a uniform size, such as 256×256 pixels. This step optimizes the batch processing and ensures uniformity across the dataset. Bilinear interpolation is used for resizing because it effectively balances computational efficiency and image quality. This method calculates the pixel value at a given location as a weighted average of the four nearest pixel values and ensures smooth transitions while preserving the critical details.

$$I'(a, b) = \sum_{p=0}^1 \sum_{q=0}^1 w_{pq} \cdot I(a_p, b_q) \quad (1)$$

where $I'(a, b)$ is the interpolated pixel value, $I(a_p, b_q)$ are the four nearest pixel values and w_{pq} represent the corresponding weights calculated on the basis of the distance from the target pixel.

Normalization is utilized to scale the pixel intensity values to a common range, typically between 0 and 1, or to adjust them to zero mean and unit variance. This process improves the statistical uniformity of the dataset and enhances the convergence rate of the neural networks during training. The normalized pixel value is calculated as follows:

$$I_{norm}(c, d) = \frac{I(c, d) - \mu}{\sigma} \quad (2)$$

where $I_{norm}(c, d)$ is the normalized pixel value, $I(c, d)$ is the actual pixel value, μ represents the average pixel intensity of the image and σ denotes the standard deviation. This ensures that the data are well suited for effective learning.

The DTCWT is utilized for noise reduction while preserving critical image details. Unlike the traditional noise-reduction methods, DTCWT provides directional sensitivity and shift invariance, enabling it to capture intricate patterns and structures. The image is decomposed into multiple sub-bands, each representing distinct frequency components and orientations.

$$I(e, f) \rightarrow \{I_{LL}, I_{LH}, I_{HL}, I_{HH}\} \quad (3)$$

where $I(e, f)$ is the input image, $I_{LL}, I_{HL}, I_{LH}, I_{HH}$ are the low-low, high-low, low-high, and high-high sub-bands, respectively. These sub-bands are further decomposed at multiple levels to extract finer details.

Following decomposition, noise reduction is applied to high-frequency sub-bands, while the low-frequency sub-band, which holds most of the structural details, is maintained. Thresholding technique such as soft or hard thresholding is applied to attenuate noise while retaining meaningful data. The thresholding operation is defined as

$$T(u) = \begin{cases} u - \lambda & \text{if } u > \lambda \\ 0 & \text{if } u \leq \lambda \\ u + \lambda & \text{if } u < -\lambda \end{cases} \quad (4)$$

where $T(u)$ is the threshold coefficient, u is the original coefficient and λ is the threshold value. The denoised image is reconstructed by applying inverse DTCWT to the modified sub-bands and ensured that the essential features

such as edges and textures are preserved. These preprocessing techniques collectively ensure that the images are optimized for subsequent analysis, enabling the reliable and accurate identification of leaf diseases. By addressing variations in the data and preserving critical features, this preprocessing pipeline provides a strong foundation for the effective deployment of deep-learning algorithms.

The Dual-Tree Complex Wavelet Preprocessing Framework (DTCWPF) delivers significant benefits compared to the traditional preprocessing methods, particularly in terms of computational efficiency and retention of critical features. Unlike the conventional approaches such as Gaussian or median filtering, which uniformly smoothen an image, DTCWPF employs dual wavelet transform trees to divide images into distinct frequency sub-bands. This hierarchical analysis allows targeted noise reduction in high-frequency regions while maintaining important low-frequency details such as structural and texture-related information. By concentrating computational efforts on relevant noise patterns and eliminating redundancies, DTCWPF achieves faster and more adaptive processing than the techniques such as K-means clustering or histogram equalization, which often requires more computational resources and lack flexibility. In terms of preserving features, DTCWPF demonstrates superior performance compared to many conventional techniques by maintaining crucial image characteristics such as textures, complex patterns, and edges which are vital indicators for detecting leaf diseases. Unlike the Gaussian blur, which tends to oversmooth and obscure fine details, DTCWPF's selective noise reduction approach ensures crucial disease-related elements, such as spots and structural anomalies, remain intact. Furthermore, the framework is extremely flexible for accommodating various conditions such as different noise levels, changes in lighting, and diverse image orientations. This adaptability ensures consistent and robust preprocessing across diverse datasets without the need of extensive parameter adjustments, unlike in many conventional preprocessing techniques.

To improve the model robustness and account for variations in real-world conditions, several data augmentation strategies have been incorporated during training and they include random flips along both axes, slight rotations, controlled zooming, and brightness adjustments. Such transformations help to increase the diversity of the training samples without altering their semantic content. Although some datasets have exhibited minor disparities in class size, stratified splitting is used to maintain proportional representation in the training, validation, and testing sets. Additionally, class balancing is handled by applying appropriate weight adjustments during training and it ensures that all categories, including those with fewer samples, have a fair influence on the model optimization. The augmentation techniques are carefully parameterized to avoid introducing

distortions that could mislead the model. Data generators are configured to apply these transformations dynamically during training and they maximize variability across epochs. As a result, the model has demonstrated improved generalization and resilience to subtle variations in unseen test samples.

3.2 Feature Extraction

EfficientNet-B7 and Inception-ResNet v2 are strategically combined for feature extraction to effectively capture vital image attributes. EfficientNet-B7 focuses on extracting high-level features, including color, texture, and intricate patterns from RGB images by utilizing efficient scaling and MBConv blocks. Concurrently, Inception-ResNet v2 processes gray-scale images by emphasizing intensity variations, edges, shapes, and structural features through its hybrid design that integrates inception modules with residual connections. The features derived from both the models are then combined to form a comprehensive feature, blending the strengths of the color and shape structural information and it greatly enhances the precision and dependability of the proposed leaf disease prediction.

3.2.1 EfficientNet-B7

Figure 2 illustrates EfficientNet-B7, a model designed to achieve high performance with a reduced number of parameters by effectively scaling across the depth, width, and resolution. This architecture uses a compound-scaling approach that adjusts these dimensions proportionally using predetermined scaling coefficients. The Mobile Inverted Bottleneck Convolution (MBConv) blocks are crucial for EfficientNet's effectiveness and they enhance the feature extraction through squeeze-and-excitation networks, depth wise separable convolutions, and skip connections. The mathematical representation of the EfficientNet-B7 model is

$$MBConv_{k \times k} = Conv_{1 \times 1} (DepthwiseConv_{k \times k} (Conv_{1 \times 1}(X))) \quad (5)$$

Here, X represents the input tensor, $DepthwiseConv_{k \times k}$ refers to depthwise separable convolution utilizing a kernel of size k , and $Conv_{1 \times 1}$ are pointwise convolutions. The squeeze-and-excitation networks use global average pooling followed by a gating mechanism to recalibrate the feature maps:

$$SE(X) = X \cdot \sigma (FC_2 (ReLU (FC_1 (GAP(X)))))) \quad (6)$$

In this context, GAP signifies global average pooling, FC_1 and FC_2 are the fully connected layers, and σ is the sigmoid activation function. This technique selectively amplifies informative features while downplaying less useful ones. The architecture of EfficientNet-B7 enables it to identify

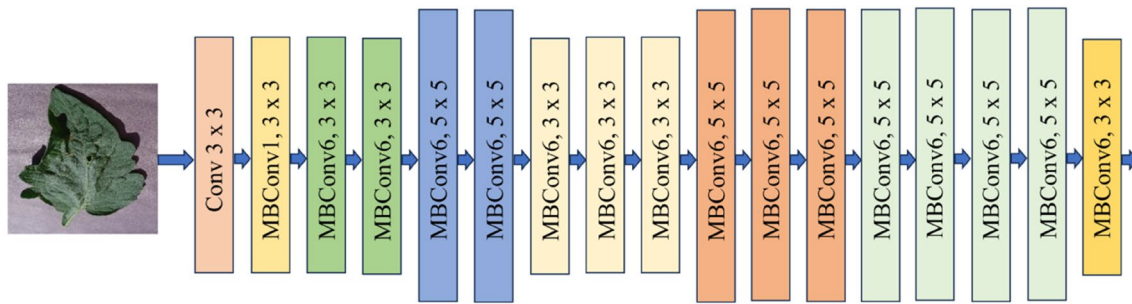


Fig. 2 Architecture of efficientNet-B7

intricate features in input images, making it highly effective for tasks such as image classification and segmentation.

EfficientNet-B7 serves as a key feature extractor in the hybrid framework, due to its proficiency in identifying complex visual patterns with remarkable efficiency. It employs a compound scaling method that evenly adjusts the network's width, depth, and resolution, enabling it to capture detailed spatial features without a significant increase in parameters. The use of inverted residual blocks and squeeze-and-excitation mechanisms helps the model with both local textures and channel interdependencies, which are especially important in capturing subtle disease symptoms such as vein discoloration or spot granularity. The model processes the input image through successive convolutional stages, in which each block progressively abstracts low-level pixel data into more descriptive and discriminative patterns. Before fusion, GAP is applied to condense the spatial dimensions into a compact feature vector that retains meaningful visual cues relevant to leaf health classification.

3.2.2 Inception-ResNet v2

Inception-ResNet v2 is a hybrid architecture that merges the benefits of inception modules with residual connections to enhance the efficiency and accuracy of the feature extraction. The Inception modules consist of multiple convolutional paths with varying filter sizes which enable the proposed algorithm to simultaneously capture features at different scales. Each Inception module is represented as follows:

$$\text{Inception}(X) = [\text{Conv}_{1 \times 1}(X), \text{Conv}_{3 \times 3}(X), \text{Conv}_{5 \times 5}(X), \text{Pooling}(X)] \quad (7)$$

The branches are merged along the channel dimensions to create the output tensor. Residual connections, as introduced by He et al., (----) facilitate the training of deeper networks by mitigating the vanishing-gradient issue and it is described as follows:

$$\text{ResNet}(X) = X + F(X) \quad (8)$$

where $F(X)$ is the residual function, typically comprising a few convolutional layers. Inception-ResNet v2 integrates these residual connections within the inception modules and it only enhances the gradient flow but also boosts the representational capacity of the network. The architecture also employs factorized convolutions, in which a convolution operation is broken into smaller, more manageable operations. This approach reduces the computational complexity while retaining the ability to capture spatial hierarchies. Inception-ResNet v2 delivers strong performance by combining efficient feature extraction and refinement in its hybrid architecture, making it a reliable solution for diverse computer vision applications.

In the context of leaf disease identification, the RGB pre-processed image is fed to EfficientNet-B7, whereas the grayscale image is processed by Inception-ResNet v2. EfficientNet-B7 extracts high-level features that encompass color information, texture, and intricate patterns. Inception-ResNet v2 focuses on the features which capture the intensity variations, edges, shapes, and structures. The combination of features from EfficientNet-B7, which uses RGB inputs, and Inception-ResNet v2, that process grayscale inputs, results in a robust feature set. This fusion capitalizes on the advantages offered by both the models.

- **Color and Texture Information:** From EfficientNet-B7, it aids in the distinction of various leaf diseases based on visual symptoms.
- **Shape and Structural Information:** From Inception-ResNet v2, it assists in the identification of the presence and extent of disease patterns.

EfficientNet-B7 and Inception-ResNet v2 are specifically selected for feature extraction because of their complementary strengths in analyzing RGB and grayscale images, thereby ensuring a comprehensive representation of critical image attributes. EfficientNet-B7 is renowned for its efficient scaling of resolution, depth, and width, enabling it to capture high-order features, such as color variations, intricate textures, and complex patterns in RGB images. The use of MBConv blocks and squeeze-and-excitation mechanisms allows it to enhance feature importance while maintaining computational efficiency. In contrast, Inception-ResNet v2

combines the multi-scale analysis capability of inception modules with the residual connections of ResNet, making it highly effective in capturing grayscale image features, such as edges, intensity gradients, shapes, and structural details. This hybrid architecture enables Inception-ResNet v2 to extract finer details from grayscale images and they are essential for detecting subtle disease patterns.

By integrating these two models, the framework has leveraged the strengths of both the architectures to extract a rich and diverse feature set. EfficientNet-B7 ensures detailed color and texture analysis, which are crucial for distinguishing visually distinct disease symptoms, while Inception-ResNet v2 provides robust shape and structural insights, which are particularly useful for identifying subtle variations in grayscale imagery. The integrated capabilities allow the framework to manage intricate datasets with varied image features, thus enhancing the accuracy and dependability of disease identification. This dual approach ensures that the critical features from both the color and grayscale domains are captured, resulting in a more effective and holistic analysis of the leaf images.

In parallel with EfficientNet-B7, Inception-ResNet v2 is utilized to extract higher-level semantic representations using a deep architecture that integrates residual connections and inception modules. The inception blocks enable multiscale feature extraction by simultaneously applying convolutions of varying kernel sizes and make the model sensitive to features at different spatial scales. This design is particularly effective in capturing structural variations, such as irregular lesion shapes or uneven edge contours. The residual connections facilitate stable gradient flow and deeper network construction, thereby enhancing learning efficiency and robustness. As the image is propagated through the network, Inception-ResNet v2 generates a high-dimensional feature map that emphasizes the contextual patterns and spatial hierarchies. These features are then processed through a global pooling layer and normalized before merging with the output of EfficientNet-B7 for ensuring that the final fused representation combines both the fine-detail resolution and the abstract semantic context.

3.3 Archimedes Optimization Algorithm (AOA) Feature Selection

The AOA used in this work for feature selection is an innovative metaheuristic approach inspired by the buoyancy principle. The AOA strikes a balance between the exploitation and the exploration within the search space by dynamically adjusting the key parameters such as volume, density, and buoyancy. This method ensures that the candidate solutions are directed towards the most promising areas of the solution space. By continuously refining these parameters and incorporating acceleration updates via a transfer function,

AOA effectively selects the most relevant features, thereby boosting the efficiency of the classifier. Its strong search capabilities contribute to higher accuracy and reliability in classification procedures, making it a critical tool in the feature-selection process of complex datasets.

It is a state-of-the-art metaheuristic optimization method inspired by Archimedes' principle of buoyancy. This concept states that the upward force exerted by a fluid on an object submerged in it is equal to the weight of the fluid displaced by the object. The AOA applies this concept by simulating buoyant forces to steer the movement of potential solutions through the search space, making it highly efficient for a variety of optimization tasks.

• Initialization and Parameters

The AOA begins with an initial population of candidate solutions and they are randomly distributed within the solution space. The initial positions of the candidates are determined using the following equation:

$$y_j(t) = y_{l,j} + rand \times (y_{u,j} - y_{l,j}) \quad (9)$$

where $y_j(t)$ denotes the position of the j -th candidate at iteration t , $y_{l,j}$ denotes the lower bounds of the solution space and $y_{u,j}$ denotes the upper bounds of the solution space for the j -th dimension. This ensures that the candidate solutions are spread across the entire search space.

• Volume, Density, and Buoyancy

AOA introduces three key parameters: volume (W), density (E), and buoyancy (Q), which are dynamically updated to effectively guide the search process. These parameters are initialized as follows:

$$W_j(t) = rand \quad (10)$$

$$E_j(t) = rand \quad (11)$$

$$Q_j(t) = y_{l,j} + rand \times (y_{u,j} - y_{l,j}) \quad (12)$$

As the algorithm is progressed, these parameters are updated using the following equations:

$$W_j(t+1) = W_j(t) + rand \times (W_{opt} - W_j(t)) \quad (13)$$

$$E_j(t+1) = E_j(t) + rand \times (E_{opt} - E_j(t)) \quad (14)$$

where W_{opt} and E_{opt} are the best volume and density values found so far and it ensures that the search process is guided towards the most promising regions within the solution space.

• Exploration and Exploitation Balance

To maintain a balance between exploitation and exploration, the AOA employs a dynamic Transfer Function (TF) and dynamic factor E :

$$TF = \frac{t_{max} - t}{t_{max}} \quad (15)$$

where TF represents a value increasing from 0 to 1, t is the current iteration, and t_{max} is the maximum number of iterations.

Factor E is adjusted to facilitate convergence.

$$E_{t+1} = \exp\left(\frac{t_{max} - t}{t_{max}}\right) - \left(\frac{t}{t_{max}}\right) \quad (16)$$

• Acceleration Updates

The acceleration of the candidate solutions is updated differently during the exploration and exploitation phases. When $TF \leq 0.5$, implying a phase of exploration with no collisions, the acceleration is updated as

$$Z_{t+1}(j) = E_{mr} + W_{mr} \times Z_{mr} E_{t+1}(j) \times W_{t+1}(j) \quad (17)$$

where Z_{mr} , E_{mr} , and W_{mr} are the acceleration, density, and volume of a randomly selected candidate.

In the exploitation phases ($TF > 0.5$), the acceleration is updated using

$$Z_{t+1}(j) = E_{opt} + W_{opt} \times Z_{opt} E_{t+1}(j) \times W_{t+1}(j) \quad (18)$$

The normalized acceleration is given by

$$Z_{t+1}(j) = u \times \frac{Z_{t+1}(j) - \min(z)}{\max(z) - \min(z)} + l \quad (19)$$

where u and l represent the normalization boundaries within $[0.9, 0.1]$.

• Position Update

Candidate positions are updated based on whether the algorithm is in the exploration or exploitation phase or not. For exploration ($TF \leq 0.5$):

$$y_{t+1}(j) = y_t(j) + c_1 \times rand \times Z_{t+1}(j) \times E \times (y_{rand} - y_t(j)) \quad (20)$$

Exploitation ($TF > 0.5$)

$$y_{t+1}(j) = y_{opt} + F \times c_2 \times rand \times Z_{t+1}(j) \times E \times (T \times y_{opt} - y_t(j)) \quad (21)$$

Where, $c_1 = 2$, $c_2 = 6$, and T ranges between $c_2 \times 0.3$ and 1.

The updated model for F is:

$$F = \begin{cases} +1, & \text{if } P \leq 0.5 \\ -1, & \text{if } P > 0.5 \end{cases} \quad (22)$$

where $P = 2 \times rand - c_4$ with ($c_4 = 0.5$).

The main objective of the AOA is to optimize the objective function g by evaluating the cost and retaining the best solution found throughout the iterations. By continuously updating the volume, density, buoyancy, and acceleration of candidate solutions, the AOA guarantees a comprehensive examination of the search space while efficiently moving towards the best solution.

The Attention-Based Quantum Convolutional Neural Network (ABQCNN) integrates quantum computing principles to improve the computational performance and feature extraction capabilities of the algorithm, particularly in complex classification tasks. By leveraging quantum operations, ABQCNN processes data in quantum bits (qubits), thereby enabling it to capture intricate patterns and relationships that are challenging for classical systems to identify. Quantum convolutional layers apply quantum gates to qubits representing image pixels, and extract essential features through entanglement and superposition principles. These layers allow the network to explore a richer feature space with fewer parameters than the classical methods and they enhance both the depth and granularity of the extracted information. Additionally, quantum pooling layers decrease the dimensionality of data while retaining essential features, thereby improving the computational efficiency without compromising the performance.

In this work, the AOA has been implemented following the quantum convolution and attention layers to identify the most informative features before classification. Candidate solutions within the algorithm are expressed as binary vectors, where each position corresponds to a feature, and a value of one signifies an inclusion. Throughout the iterative process, these candidate subsets are evaluated based on their contributions to classification accuracy, enabling direct alignment between the selected features and the task performance. The use of dynamically evolving physical parameters, namely, volume, density, and buoyancy, enables the optimizer to escape from local optima and guide the search toward more promising feature combinations. This makes AOA particularly effective in scenarios involving high-dimensional nonlinear data. Unlike static or filter-based selection methods, this physics-inspired approach balances global search capabilities with convergence speed, ensuring that the resulting feature set maximizes both the relevance and the compactness for downstream learning.

Pseudo-Code for AOA based feature selection

Step 1: Initialization

Initialize Population of Candidate Solutions: Population = {y1, y2, ..., yn}

Set Maximum Iterations: t_max

Set Parameters: W_opt, E_opt, c1 = 2, c2 = 6, T ranges between c2 * 0.3 and 1, u = 0.9, l = 0.1

Initialize Volume (W), Density (E), Buoyancy (Q) for each candidate

Initialize the Best Solution: y_opt

Step 2: Calculate Initial Positions of Candidates

for each candidate j in the population.

$y_j(t) = y_{l,j} + \text{rand}() * (y_{u,j} - y_{l,j})$

Step 3: Fitness Calculation

for each candidate j in the population.

Calculate Fitness(y_j(t)) = g(objective function)

Step 4: Update Best Solution

Update y_opt if a better solution is found

Step 5: Main Loop

for iteration t in the range of (1, t_max + 1).

Step 6: Update Volume, Density, and Buoyancy

for each candidate j in the population.

$W_j(t+1) = W_j(t) + \text{rand}() * (W_{\text{opt}} - W_j(t))$

$E_j(t+1) = E_j(t) + \text{rand}() * (E_{\text{opt}} - E_j(t))$

$Q_j(t+1) = y_{l,j} + \text{rand}() * (y_{u,j} - y_{l,j})$

Step 7: Calculate Transfer Function (TF)

$TF = (t_{\text{max}} - t) / t_{\text{max}}$

Step 8: Update Acceleration based on TF

if $TF \leq 0.5$: # Exploration phase

for each candidate j in the population.

$Z_{(t+1)}(j) = E_{\text{mr}} + W_{\text{mr}} * Z_{\text{mr}} * E_{(t+1)}(j) * W_{(t+1)}(j)$

else: # Exploitation phase

for each candidate j in the population.

$Z_{(t+1)}(j) = E_{\text{opt}} + W_{\text{opt}} * Z_{\text{opt}} * E_{(t+1)}(j) * W_{(t+1)}(j)$

Normalize Acceleration

for each candidate j in the population.

$Z_{(t+1)}(j) = u * (Z_{(t+1)}(j) - \min(Z)) / (\max(Z) - \min(Z)) + l$

```

# Step 9: Update Candidate Positions based on TF
for each candidate j in the population.
    if  $TF \leq 0.5$ : # Exploration phase
         $y_{(t+1)}(j) = y_{(t)}(j) + c1 * rand() * Z_{(t+1)}(j) * E * (y_{rand} - y_{(t)}(j))$ 
    else: # Exploitation phase
         $F = +1$  if  $rand() \leq 0.5$  else  $-1$ 
         $P = 2 * rand() - c4$  #  $c4 = 0.5$ 
         $y_{(t+1)}(j) = y_{opt} + F * c2 * rand() * Z_{(t+1)}(j) * E * (T * y_{opt} - y_{(t)}(j))$ 

# Step 10: Fitness calculation
for each candidate j in the population.
    Calculate  $Fitness(y_j(t+1)) = g$  (objective function)

# Step 11: Update Best Solution
Update  $y_{opt}$  if a better solution is found

# Step 12: Convergence Check
if iteration  $> t_{max}$  or  $|Fitness(y_{opt}) - previous\_best\_fitness| < \epsilon$ .
    break # Convergence condition met

# Update previous_best_fitness with the current best fitness

# Step 13: Output the Best Solution
Return Best Solution  $y_{opt}$  and its corresponding feature subset

# End of Algorithm

```

3.4 Attention Based Quantum Convolutional Neural Networks - Classification

The ABQCNN, as shown in Fig. 3, has been utilized in this study for classification and it integrates quantum computing principles with classical CNNs to enhance feature extraction and classification accuracy. By performing convolutional operations on quantum bits (qubits), complex feature interactions are captured more efficiently, whereas

quantum-pooling layers reduce the data dimensionality and retain critical features. By incorporating an attention mechanism, the model enhances its focus on the most pertinent information and thereby, its capacity is improved to emphasize essential features. This combination of quantum computing and attention mechanisms is a powerful and an effective tool for achieving high accuracy in complex classification tasks.

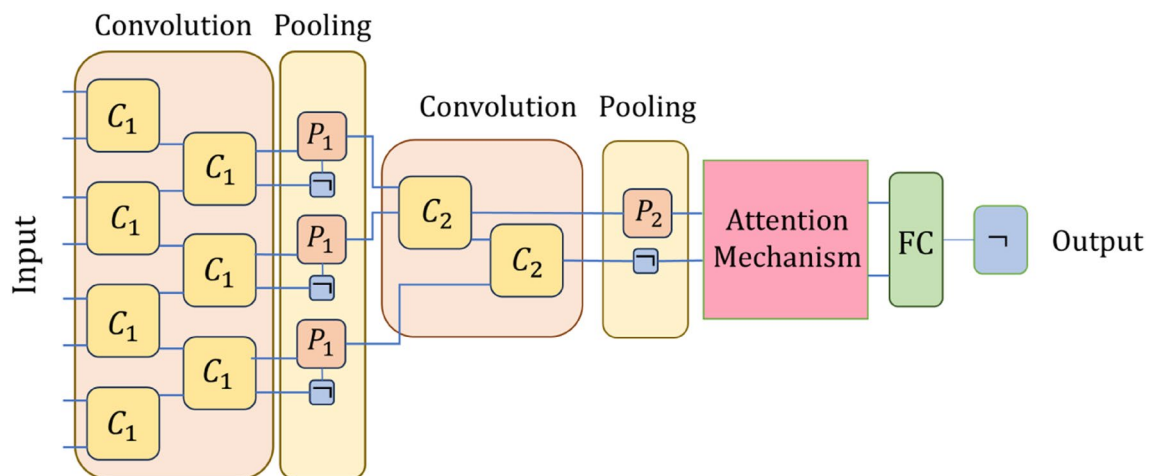


Fig. 3 Structure of the proposed ABQCNN

QCNNs extend classical CNNs by incorporating quantum computing principles. In a QCNN, convolutional operations are performed on quantum bits (qubits) instead of on classical bits. This approach allows the capture of complex feature interactions in a computationally efficient manner. Quantum convolutional layers apply quantum gates to qubits, which represent image pixels, to extract essential features. The quantum convolution operation is defined as follows.

$$QCNN_{conv}(a, b) = \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} w_{pq} \cdot QCNN(a_p, b_q) \quad (23)$$

where:

- $QCNN_{conv}(a, b)$ is the convolved quantum pixel value,
- $QCNN(a_p, b_q)$ represents the qubits corresponding to the image pixels,
- w_{pq} are quantum convolution weights.

Quantum convolution involves the application of quantum gates to qubits, mirroring the classical convolution process, and selecting the features from an input image. Quantum pooling layers decrease data dimensionality while preserving the most important features. This step helps to reduce computational complexity and emphasizes important information. The pooling operation is applied to quantum feature maps to maintain critical features while minimizing noise. The quantum pooling operation can be represented as

$$QCNN_{pool}(a, b) = \frac{1}{k^2} \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} QCNN_{conv}(a_p, b_q) \quad (24)$$

After pooling, the FC quantum layers integrate the features extracted by the pooling and convolutional layers. These layers help to classify images based on aggregated features. The quantum FC layer can be represented as

$$QCNN_{fc} = \sigma \left(\sum_i w_i \cdot QCNN_{pool}(a_i, b_i) + b \right) \quad (25)$$

where:

- $QCNN_{fc}$ is the output of the FC layer,
- w_i is the FC layer weights,
- b denotes the bias term,
- σ is the activation function (ReLU).

The attention mechanism is incorporated with QCNN to enhance its capability to focus on the most essential features. By assigning varying levels of importance to different features, the network can focus on the most critical information. The attention weights are calculated using a mechanism that evaluates the significance of each feature. The attention mechanism is represented by

$$Attention(X) = \sigma(W_2 \cdot ReLU(W_1 \cdot X + b_1) + b_2) \quad (26)$$

where:

- X is the input feature map,
- W_1 and W_2 are the weight matrices,
- b_1 and b_2 are the bias terms,
- σ is the sigmoid activation function.

By leveraging quantum operations, data are processed in quantum bits (qubits), thereby enabling to capture intricate patterns and relationships that are challenging for classical systems to identify. Quantum convolutional layers apply quantum gates to qubits representing image pixels, and extract essential features through entanglement and superposition principles. These layers allow the network to explore a richer feature space with fewer parameters than the classical methods, enhancing both the depth and the granularity of the extracted information. Additionally, quantum pooling layers decrease the data dimensionality while retaining essential features, thereby improving the computational performance without compromising the performance. The incorporation of an attention mechanism is further strengthened by dynamically prioritizing the most relevant features during training. The attention mechanism prioritizes significant features by assigning them with greater weights and allows the system to concentrate on essential regions of the image, while ignoring less crucial details. This synergistic combination of quantum computation and attention mechanisms significantly improves the model's ability to identify subtle patterns and optimizes the feature representation. As a result, it delivers higher classification accuracy, faster convergence, and superior generalization, particularly for the datasets that require the analysis of complex and diverse features. These advantages make it a powerful tool for addressing challenges in computationally demanding tasks, such as agricultural disease detection.

Pseudo-Code for ABQCNN based classification

```

# Initialize quantum CNN parameters
Initialize Quantum Convolution Weights: W_conv
Initialize Quantum Pooling Weights: W_pool
Initialize Fully Connected Layer Weights: W_fc, Bias: b_fc
Initialize Attention Mechanism Weights: W1, W2, Bias: b1, b2

# Input: Quantum image represented by qubits
Input: Quantum Image Q

# Step 1: Quantum Convolutional Layer
for each quantum pixel (a, b) in Quantum Image Q:
  Convolved_Quantum_Pixel = 0
  for p = 0 to k-1:
    for q = 0 to k-1:
      Convolved_Quantum_Pixel += W_conv[p, q] * QCNN(a_p, b_q)
  QCNN_conv(a, b) = Convolved_Quantum_Pixel

# Step 2: Quantum Pooling Layer
for each quantum feature map (a, b) in the Convolved Quantum Image
  Pooled_Quantum_Pixel = 1/k^2 * sum (QCNN_conv(a_p, b_q) for p in range(k) for q in range(k))
  QCNN_pool(a, b) = Pooled_Quantum_Pixel

# Step 3: Attention Mechanism
for each quantum feature map X:
  Attention_Weights =  $\sigma(W2 * \text{ReLU}(W1 * X + b1) + b2)$ 
  Attention_Enhanced_Features = Attention_Weights * QCNN_pool(a, b)

# Step 4: Quantum FC Layer
for each attention-enhanced feature:
  QCNN_fc_output =  $\sigma(\text{sum}(W\_fc[i] * \text{Attention\_Enhanced\_Features}(a\_i, b\_i) + b\_fc \text{ for } i \text{ in range}(\text{len}(\text{Attention\_Enhanced\_Features}))))$ 

# Step 5: Classification Output
Output: Classify image based on QCNN_fc_output

# End of Algorithm

```

The combination of quantum principles with the CNN is motivated by the ability of quantum-inspired operations to capture richer feature representations with fewer computational resources. In the proposed model, the quantum convolution layer substitutes the conventional filters with unitary transformations that simulate quantum behavior, thereby enabling entangled and superposed feature mappings. This allows the network to encode relationships across the feature space in a non-classical manner and sensitivity is enhanced to subtle variations which often present in leaf disease patterns. Additionally, the introduced quantum attention mechanism acts as a probabilistic weighting scheme, where

feature relevance is modulated through phase-inspired interference and the focus on disease-affected regions is improved without introducing excessive parameter overhead. Unlike the classical CNNs, which rely solely on local spatial hierarchies, the quantum-enhanced approach leverages mathematically distinct operations to uncover complex dependencies, which contribute to improved classification accuracy. This integration provides not just an architectural variation, but also a meaningful theoretical shift that augments the model's ability to generalize, particularly in fine-grained agricultural datasets.

3.5 Adaptive Silver Fox Algorithm for Classification

The Adaptive Silver Fox Algorithm (ASFA) utilized in this work for classification is an advanced feature selection method inspired by the adaptive hunting strategies of silver foxes. By dynamically balancing exploration and exploitation, ASFA effectively identifies the most essential features from a dataset and enhances the efficiency of classification models in complex procedures, such as leaf disease identification. The algorithm iteratively refines candidate solutions, adjusts search parameters based on environmental feedback, and ensures diversity through selection, crossover, and mutation processes. This adaptive approach allows ASFA to converge on optimal feature subsets, resulting in more reliable and accurate classification outcomes. The ASFA is an advanced feature selection method inspired by the hunting strategies and adaptive behavior of silver foxes. This algorithm aims to select the most relevant features from a dataset to improve the performance of machine learning models, particularly in complex tasks, such as leaf disease identification.

• Fox Population Initialization

The algorithm starts by creating a population of candidate solutions and each of which represents a potential subset of features. These candidates are analogous to a group of silver foxes in the search space.

$$Population = \{F_1, F_2, \dots, F_n\} \quad (27)$$

where F_i represents a subset of features.

• Fitness Evaluation:

Each candidate solution (fox) is evaluated using a fitness function, which typically measures the efficiency of a classifier model based on the selected features. Metrics such as classification accuracy, precision, recall, or a combination these can be used to determine the fitness.

$$Fitness(F_i) = g(model\ performance\ on\ F_i) \quad (28)$$

• Hunting Strategy and Exploration

Foxes employ a hunting strategy that combines exploration with exploitation. Exploration allows the algorithm to search broadly across the feature space to identify the promising regions, whereas exploitation focuses on refining solutions in areas with high potential. The adaptive behavior of silver foxes is modeled through a balance between local search (exploring nearby features) and global search (exploring diverse feature subsets).

Exploration:

$$F'_i = F_i + \alpha \cdot rand() \quad (29)$$

Exploitation:

$$F'_i = F_i + \beta \cdot (best\ solution - F_i) \quad (30)$$

• Adaptive Behaviour:

The adaptive component of the ASFA enables foxes to adjust their search patterns based on the environmental feedback. This includes dynamically tuning parameters, such as the search radius and step size, allowing the algorithm to adapt to the complexity and structure of the feature space.

$$\alpha, \beta \propto \frac{1}{1 + e^{-\gamma (current\ iteration - \frac{maxiterations}{2})}} \quad (31)$$

• Selection and Reproduction

This algorithm selects the most promising candidates based on their fitness scores. These selected foxes undergo reproduction mechanisms such as crossover and mutation to generate new candidate solutions. This process ensures population diversity and prevents premature convergence.

Crossover:

$$F_{new} = \frac{F_i + F_j}{2} \quad (32)$$

Mutation:

$$F_{new} = F_i + \delta \cdot rand() \quad (33)$$

• Convergence Criteria:

The algorithm iteratively updates the population of the foxes by continuously improving the quality of the selected feature subsets. Convergence is generally recognized by either completing a predetermined number of iterations or when the gains in fitness scores decrease to a certain threshold.

$$Convergence = \begin{cases} True & \text{if iteration count exceeds max iterations} \\ True & \text{if } \Delta Fitness < \epsilon \\ False & \text{otherwise} \end{cases} \quad (34)$$

The ASFA enhances the model's classification performance by employing an adaptive strategy inspired by the hunting behavior of silver foxes as well balances exploration and exploitation within the feature space. This dynamic optimization strategy ensures that the model pinpoints the most significant features for classification, while simultaneously preventing overfitting and avoiding convergence to sub-optimal outcomes. ASFA achieves this by adaptively refining the candidate solutions based on the feedback from the environment and dynamically adjusting parameters, such as the search radius and step size. This adaptability allows the algorithm to efficiently navigate complex feature spaces by ensuring thorough exploration during the initial phases, and focuses exploitation as the search progresses towards an optimal solution. Key factors in the design of ASFA include

its mechanisms for maintaining diversity in candidate solutions and its ability to adaptively balance the trade-off between global exploration and local exploitation. Techniques, such as mutations and crossover mutation, ensure population diversity, prevent premature convergence, and

enable the technique to explore a broader solution space. The inclusion of adaptive behavior, guided by transfer functions and dynamic parameter adjustments, ensures that the algorithm can respond effectively to varying complexities in datasets.

Pseudo-Code for Adaptive Silver Fox Algorithm (ASFA)

```
# Step 1: Initialization
Initialize Population of Foxes: Population = {F1, F2, ..., Fn}
Initialize Parameters:  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ 
Set Maximum Iterations: max_iterations
Set Convergence Threshold:  $\epsilon$ 

# Step 2: Fitness Evaluation
for each Fox Fi in the Population:
    Calculate Fitness (Fi) = g(model performance on Fi)

# Step 3: Main Loop
for iterations within the range (1, max_iterations + 1).
    # Step 4: Hunting Strategy (Exploration and Exploitation)
    for each Fox Fi in the Population:
        # Exploration
        Fi_new = Fi +  $\alpha$  * random ()
        # Exploitation (Move towards the best solution found so far)
        Fi_new = Fi +  $\beta$  * (Best_Fox - Fi)

        # Update Fi if Fi_new is better
        if fitness (Fi_new) > fitness (Fi).
            Fi = Fi_new

    # Step 5: Adaptive Behavior (Adjust  $\alpha$  and  $\beta$ )
     $\alpha = 1 / (1 + \exp (-\gamma * (\text{iteration} - \text{max\_iterations} / 2)))$ 
     $\beta = 1 / (1 + \exp (-\gamma * (\text{iteration} - \text{max\_iterations} / 2)))$ 

# Step 6: Selection and Reproduction
Select the top-performing Foxes based on Fitness Scores

for i in range (1, len(Population)/2): # Assume half population reproduces
    # Crossover
    Fj = Select another Fox for crossover
    F_new = (Fi + Fj) / 2
    # Mutation
    if random () < mutation_rate: # mutation_rate is a predefined probability
        F_new = F_new +  $\delta$  * random ()
    # Add F_new to Population

# Step 7: Update Best Solution
Update Best_Fox if a better solution is found in this iteration

# Step 8: Convergence Check
if iteration > max_iterations or abs (fitness (Best_Fox) - previous_best_fitness) <  $\epsilon$ 
    break # Convergence condition met
# Update previous_best_fitness with the current Best_Fox fitness

# Step 9: Output the Best Solution
Return Best_Fox and its corresponding feature subset
# End of Algorithm
```

The notable accuracy achieved by the proposed algorithm can be attributed to the synergistic effect of several carefully integrated components, each of which contributes to different aspects of the learning pipeline. One key factor is DTCWPF which improves the image quality by isolating directional features and suppressing irrelevant noise while preserving critical texture and edge information. This pre-processing strategy ensures that the model receives input with improved clarity and structure and allows the subsequent layers to extract more meaningful patterns related to disease-specific symptoms. Another significant contributor is AOA which plays a vital role in refining the feature space before classification. By evaluating and selecting only the most relevant features from the output of the quantum convolutional and attention modules, the AOA effectively reduces redundancy and improves the discriminative power of the model. This tailored selection not only minimizes overfitting, but also accelerates convergence by focusing on information that directly correlates with class-specific patterns.

Furthermore, the integration of quantum-inspired layers has introduced a new dimension of representational power. Quantum convolution mimics superposition and

entanglement principles to extract richer relationships between the pixels that might be overlooked by classical kernels. Coupled with a quantum-guided attention mechanism, the model can prioritize subtle yet informative regions of the image, such as minor discolorations or irregular margins. These layers collectively contribute to the model's finely tuned sensitivity and specificity, especially in distinguishing visually similar disease categories. Together, these architectural choices form a coherent framework that strengthens the model's ability to generalize across varied datasets while maintaining high predictive accuracy.

The integration of both the Archimedes Optimization Algorithm (AOA) and Adaptive Softmax Fusion Algorithm (ASFA) within a single pipeline is a strategic design aimed at addressing the distinct and complementary stages of the classification framework. The AOA is specifically employed for feature selection, where its physics-inspired search mechanism efficiently identifies the most relevant and nonredundant features from high-dimensional space generated by dual-path deep networks. This step is critical for minimizing computational overhead and enhancing the discriminative power of the input to the classifier.

Cotton Leaf Dataset 1

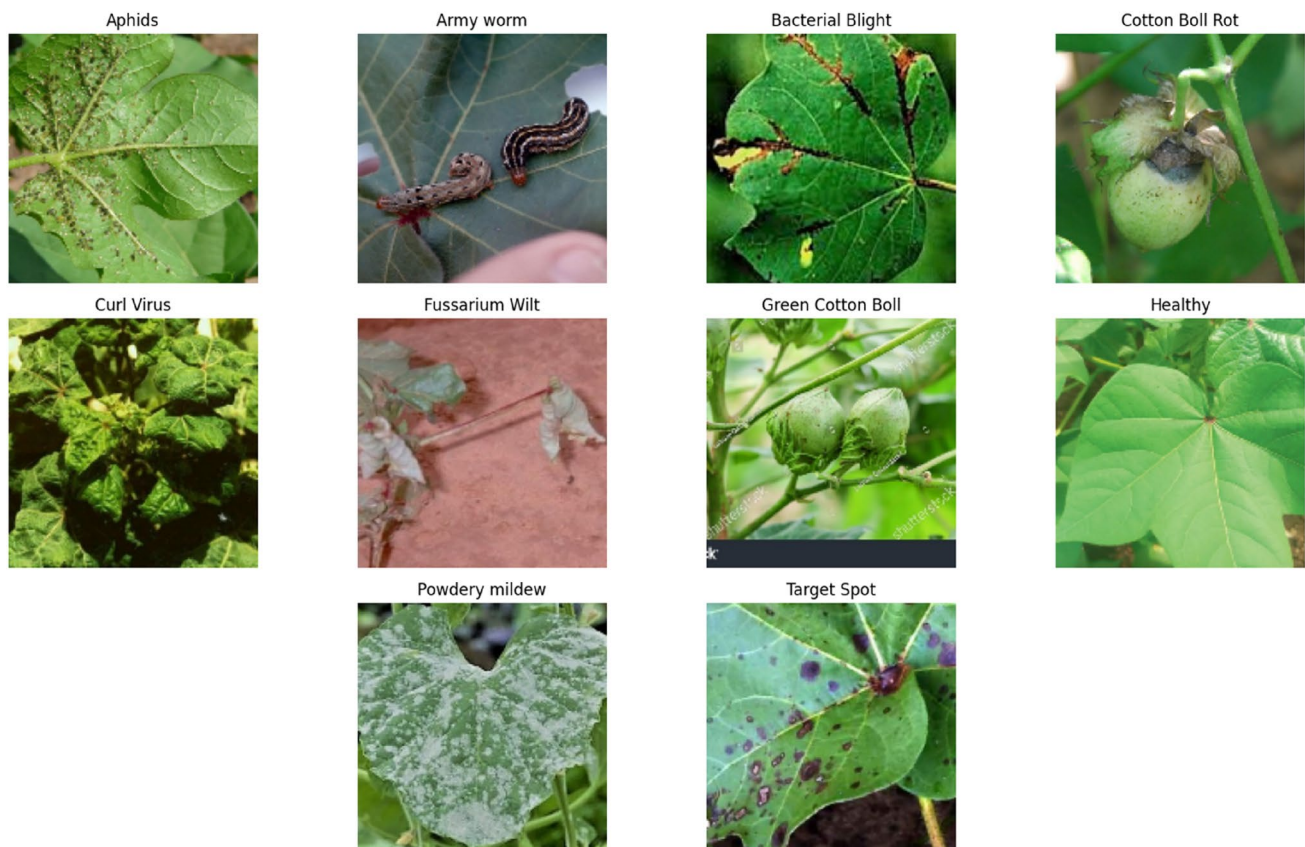


Fig. 4 Cotton leaf dataset 1

In contrast, ASFA is introduced at the classification level to refine the decision boundaries through adaptive weight fusion of multiple classifier outputs. It dynamically learns to emphasize more informative predictions and suppresses noise or weak signals in ensemble outputs. Hence, it is particularly beneficial in cases of overlapping class features or subtle disease symptoms. Using only one of these components would limit either the optimization of feature relevance or the robustness of the final classification. Consequently, their combination is not redundant but rather synergistic. The AOA enhances the quality of features entering the classification stage, whereas ASFA optimally fuses classifier responses to boost the final prediction accuracy. This layered design contributes to overall performance gains and justifies the added complexity.

4 Result and Discussion

The efficiency of the proposed deep learning framework in identifying leaf diseases has been assessed using various datasets, including those for cotton and tomato leaves. This proposed work examines the model's precision, accuracy, specificity, and other critical metrics and its effectiveness is compared with the existing models as well as its capability is explored to generalize and accurately classify a range of plant diseases. This section also includes a detailed examination of the model's training and validation processes by

highlighting its superior performance in agricultural disease detection.

4.1 Dataset

Cotton Dataset 1: Cotton Leaf Dataset 1 has been collected from a publicly available source on Kaggle (<https://www.kaggle.com/datasets/asrafulsharkernirob/cotton-final-leaf>) and it includes 9,463 RGB images. The data span 10 well-defined categories: Aphids (787), Army Worm (799), Bacterial Blight (1245), Cotton Boll Rot (960), Curl Virus (418), Fusarium Wilt (411), Green Cotton Boll (880), Powdery Mildew (800), Target Spot (788), and Healthy leaves (1225). All images are organized into class-specific directories and visually consistent in format. Since no original split for training, validation, or testing is provided, a stratified sampling approach is applied by allocating 80% of the data for model training, 10% for validation, and 10% for final evaluation, which ensure balanced representation from each disease class throughout the pipeline. Figure 4 shows Cotton Leaf Dataset 1.

Cotton Dataset 2: Cotton Leaf Dataset 2 has been obtained from an open-source repository on Kaggle (<http://www.kaggle.com/datasets/asrafulsharkernirob/cotton-leaf-augmented-7-class>) comprising 7,000 images and it is equally distributed across seven categories, each consisting of 1,000 labelled samples. The included classes are Bacterial Blight of Cotton, Curl Virus, Healthy Leaf, Herbicide Growth Damage, Leaf Hopper Jassids, Leaf Redding, and Leaf Variegation.

Cotton Leaf Dataset 2

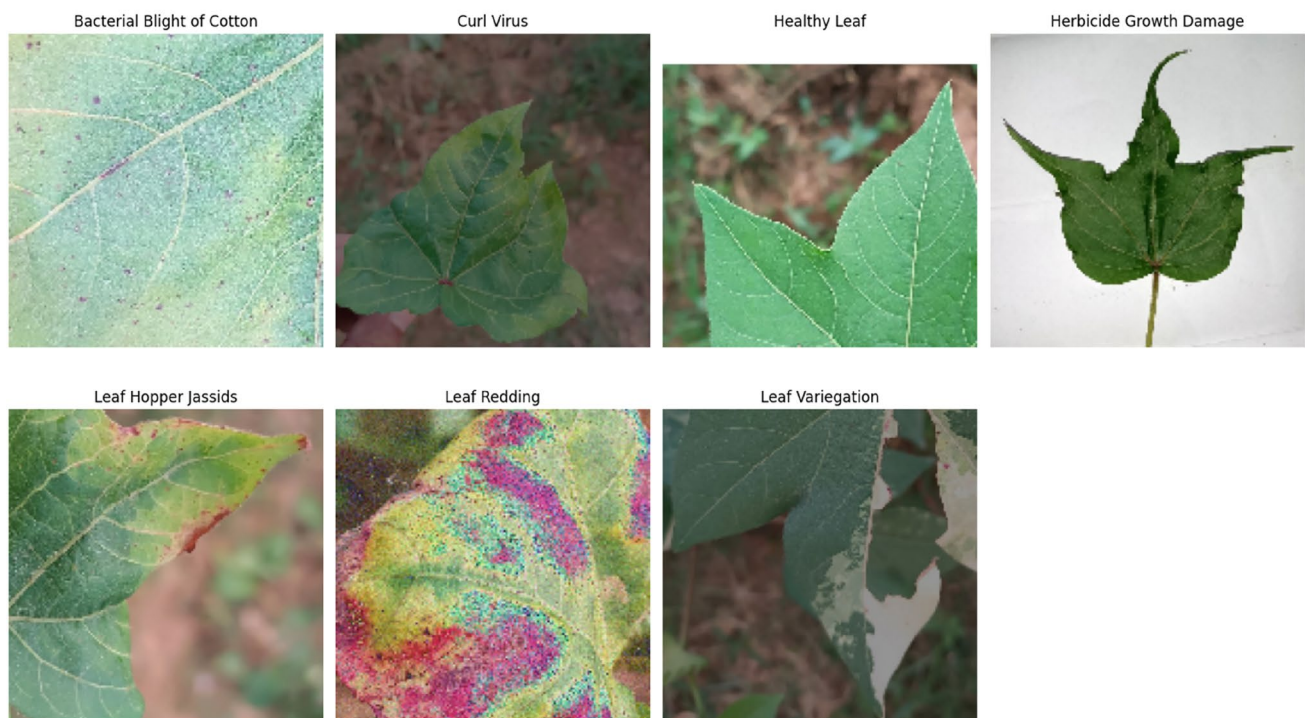


Fig. 5 Cotton leaf dataset 2

Tomato Dataset 1

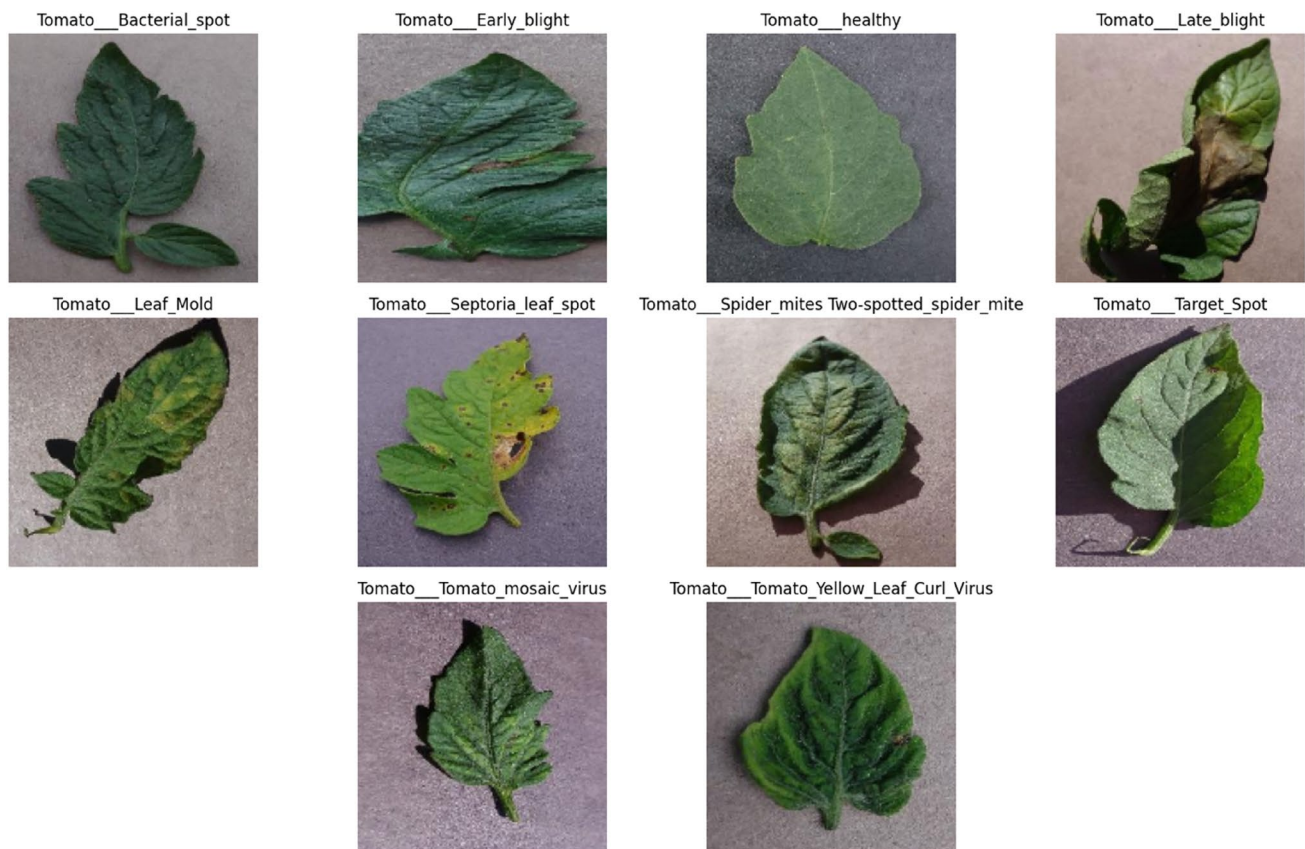


Fig. 6 Tomato leaf dataset 1

Table 2 Experimental configuration details

Component	Details
CPU	Intel Core i9-11900 K, 8 cores, 3.5 GHz
GPU	NVIDIA RTX 3080, 10 GB GDDR6X
RAM	64 GB DDR4
Operating System	Ubuntu 20.04 LTS (64-bit)
CUDA Toolkit	CUDA 11.2
cuDNN	cuDNN 8.1
Python Version	Python 3.8
Deep Learning Frameworks	TensorFlow 2.6, PyTorch 1.9
Key Libraries	NumPy, SciPy, OpenCV

Table 3 Experimental parameters

Parameters	Value
Batch size	16
Loss	Categorical Cross entropy
Optimizer	Adam
Learning rate	0.0001
Activation	Softmax
Dropout	0.5
Epoch	30

Growth Damage, Leaf Hopper Jassids, Leaf Redding, and Leaf Variegation. All the images are augmented and uniformly structured in terms of resolution and visual clarity. Since the dataset does not include pre-arranged subsets, the images are classified into 80%, 10%, and 10% for training, validation, and testing, respectively. Figure 5 depicts Cotton Leaf dataset-2.

Tomato Dataset 1: Tomato Leaf Dataset 1 has been obtained from the PlantVillage collection hosted on Kaggle (<https://www.kaggle.com/datasets/adilmubashirchaudhry/plant-village-dataset>). It contains more than 18,000 labelled images, grouped into ten tomato-related classes: Bacterial Spot (2127), Early Blight (1000), Late Blight (1909), Leaf Mold (952), Septoria Leaf Spot (1771), Spider Mites (1676), Target Spot (1404), Tomato Yellow Leaf Curl Virus (3209), Tomato Mosaic Virus (373), and Healthy (1591). The dataset includes only tomato categories with uniform RGB images suitable for deep-learning applications. Since no train-test split is originally defined, a stratified division is applied to maintain class balance by assigning 80% of the data for training, 10% for validation, and 10% for testing. Figure 6 demonstrates tomato leaf dataset-1.



Fig. 7 Confusion matrix of tomato dataset 1

4.2 Experimental Configuration

This experimental setup is featured with a medium-configuration system tailored for intensive machine learning and deep learning workloads. Table 2 presents the details of the experimental configuration. The hardware setup has included an Intel Core i9-11900 K processor, which has eight cores running at 3.5 GHz, and an NVIDIA RTX 3080 GPU with 10 GB of GDDR6X memory. The system is also equipped with 64 GB of DDR4 RAM, ensuring sufficient capacity for multitasking and processing large datasets. In terms of software, the system operates on Ubuntu 20.04 LTS (64-bit) and leverages CUDA 11.2 alongside cuDNN 8.1 for GPU acceleration in deep learning tasks. Python 3.8 with 2.6 serves as the main programming language, which is available for model development and training. Essential libraries, such as NumPy, SciPy, and OpenCV, are also installed for data processing and image handling

and a comprehensive environment is created for conducting sophisticated machine learning experiments.

4.3 Experimental Parameters

A neural network model has been developed using a carefully selected set of hyperparameters to achieve strong performance and generalization. As shown in Table 3, a batch size of 16 is used to strike a balance between the computational efficiency and the convergence by allowing frequent weight updates while ensuring model stability. The loss function used is Categorical Cross Entropy, which is ideal for multiclass classification problems and it effectively addresses any incorrect predictions made by the model. The Adam optimizer, chosen for its adaptive learning rate capabilities, is used to promote faster convergence with a learning rate of 0.0001 to ensure that the learning process is both gradual and stable. The output layer employs the Softmax activation function, which converts the model outputs into probabilities for each class, thereby facilitating accurate classification. To combat overfitting, a dropout rate of 0.5 is introduced for randomly deactivating neurons during training as well as to encourage the learning of more generalized features. Training has been conducted over 30 epochs by providing sufficient time for the model to learn the data patterns without overfitting, as demonstrated by the resulting performance metrics.

4.4 Experimental Results

4.4.1 Tomato Dataset 1 Analysis

Figure 7 presents the confusion matrix for Tomato Dataset 1 and it demonstrates the model's strong capability in accurately identifying various tomato leaf diseases. It shows perfect classification for certain categories, such as Bacterial

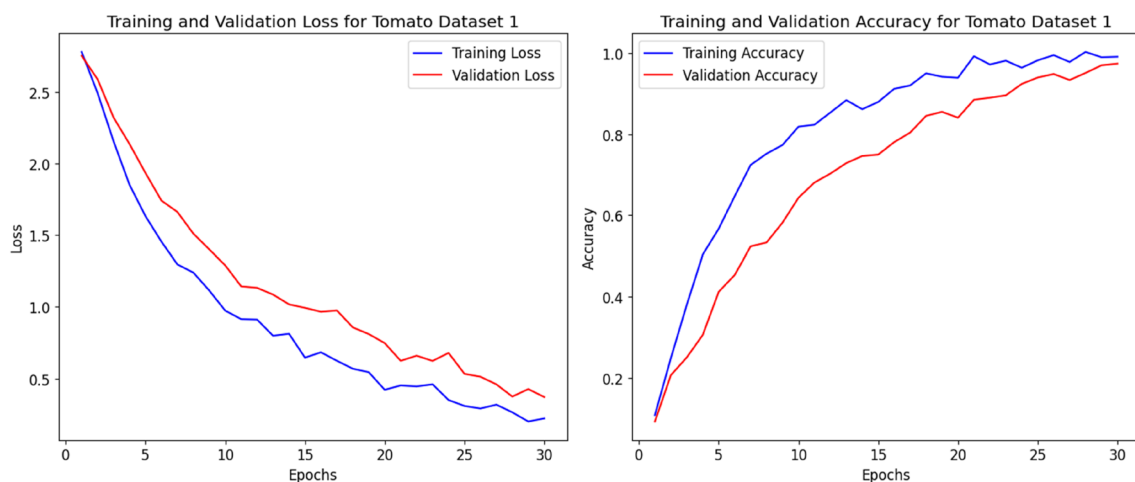


Fig. 8 Training and validation of tomato dataset 1

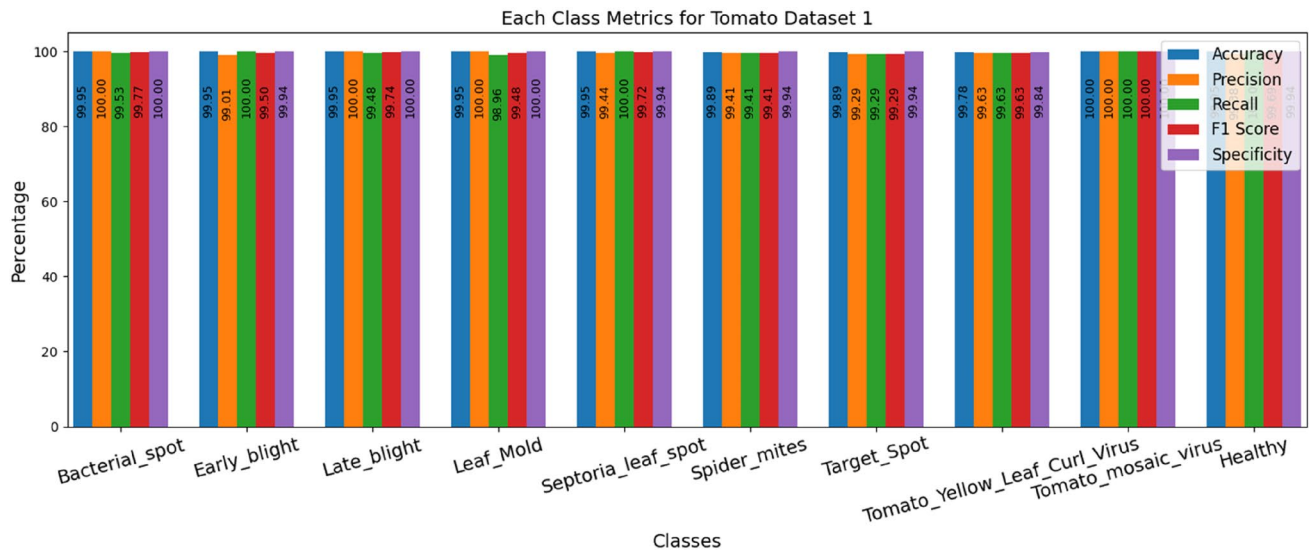


Fig. 9 Performance metrics of tomato dataset 1

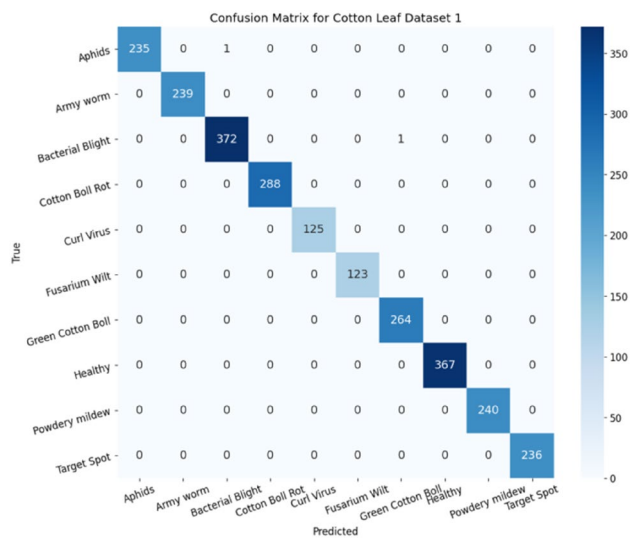


Fig. 10 Confusion matrix of cotton leaf dataset 1

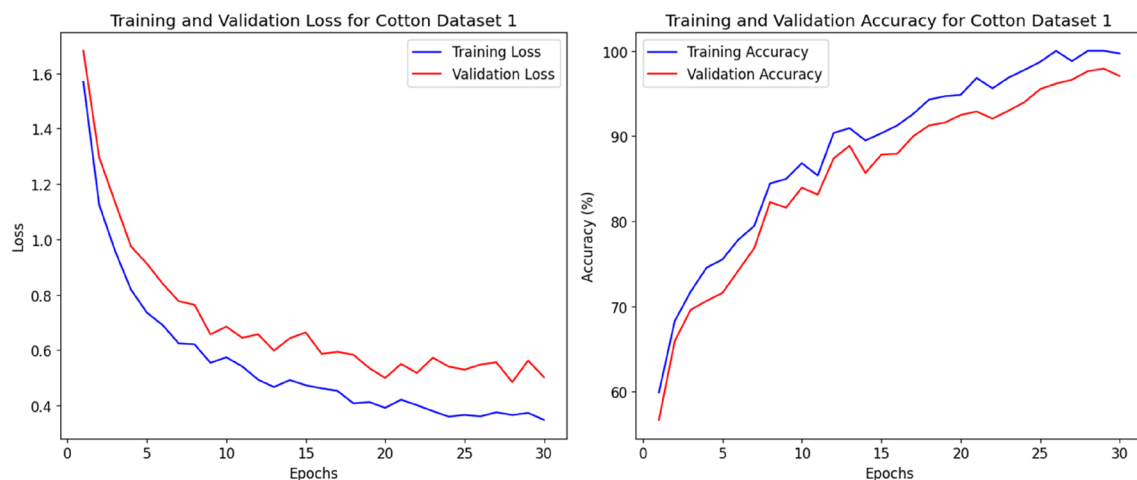


Fig. 11 Training and validation of cotton leaf dataset 1

Spot, Early Blight, and Healthy, with no errors detected. A few misclassifications are observed, such as Late Blight being wrongly classified as Spider Mites, but the overall accuracy remains remarkably high. This level of performance reflects the effectiveness and reliability of the model in detecting various tomato leaf diseases by making it a valuable asset for agricultural disease detection. The training and validation curves portrayed in Fig. 8 for Tomato Dataset 1 over 30 epochs demonstrate a well-balanced learning process. The loss consistently decreases, with the training loss enduring slightly below the validation loss, indicating effective learning with minimal overfitting. The accuracies of both training and validation improve steadily and remain closely aligned, suggesting that the model effectively learns the data patterns and is likely to perform well on new, unseen data. This alignment of training and

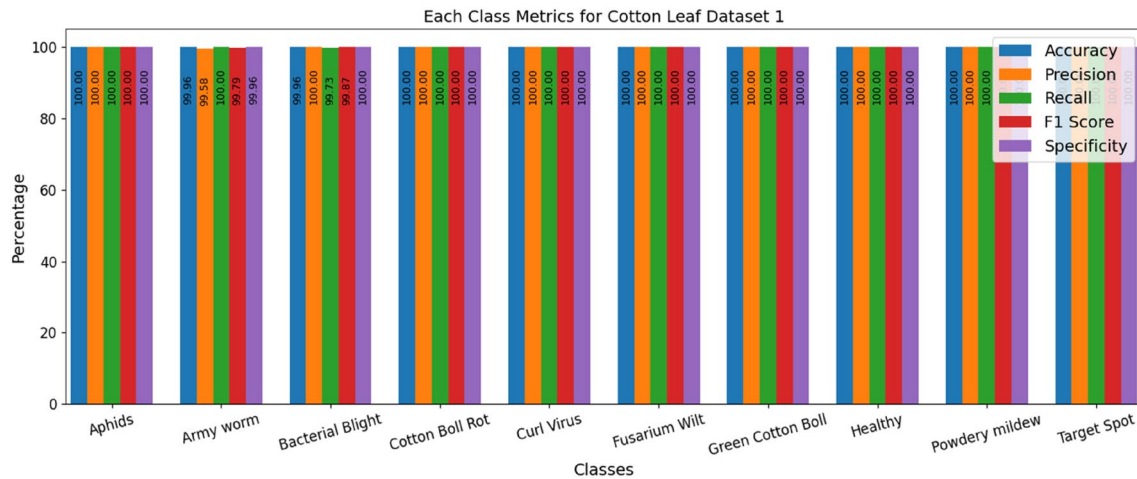


Fig. 12 Performance metrics of cotton leaf dataset 1

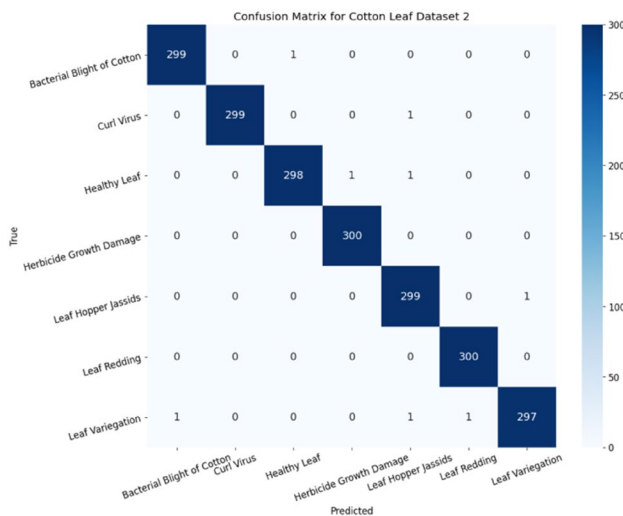


Fig. 13 Confusion matrix of cotton leaf dataset 2

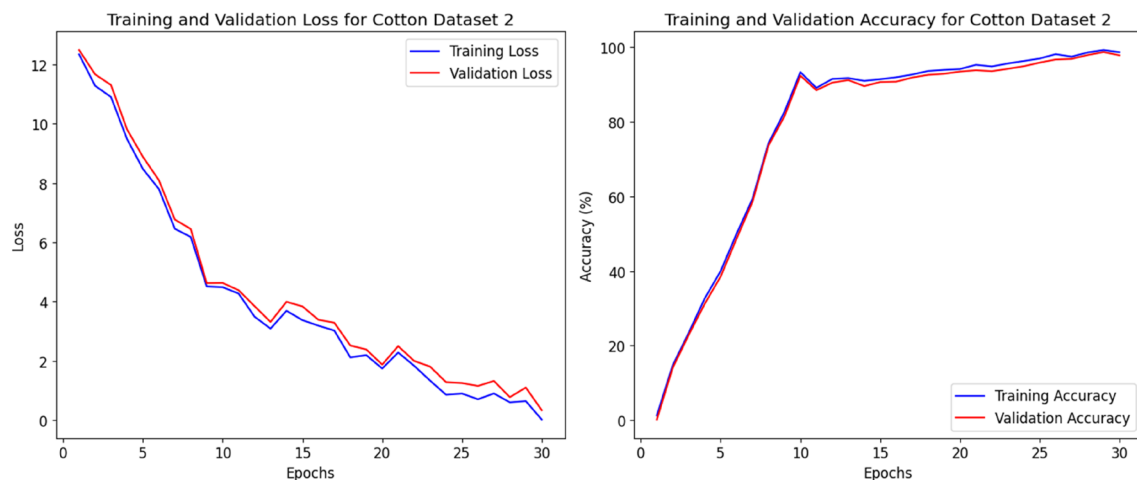


Fig. 14 Training and validation of cotton leaf dataset 2

validation metrics demonstrates the model's capability to generalize and accurately classify tomato leaf diseases.

The bar chart shown in Fig. 9, depicting class-specific metrics for Tomato Dataset 1, highlights excellent performance of the model across all classes. Metrics such as accuracy, precision, recall, F1 score, and specificity are nearly perfect for each category and they reflect the model's consistent and reliable predictions. This uniformity across the metrics highlights the model's capability to accurately classify with few mistakes, establishing it as a highly reliable tool for diagnosing tomato leaf diseases.

4.4.2 Cotton Leaf Dataset 1 Analysis

The confusion matrix of Cotton Leaf Dataset 1 is shown in Fig. 10 and it emphasizes the strong ability of the model to correctly classify various cotton leaf conditions. The majority of correct predictions falls along the diagonal, with classes such as Bacterial Blight, Healthy, and Powdery Mildew showing high accuracy. The few misclassifications

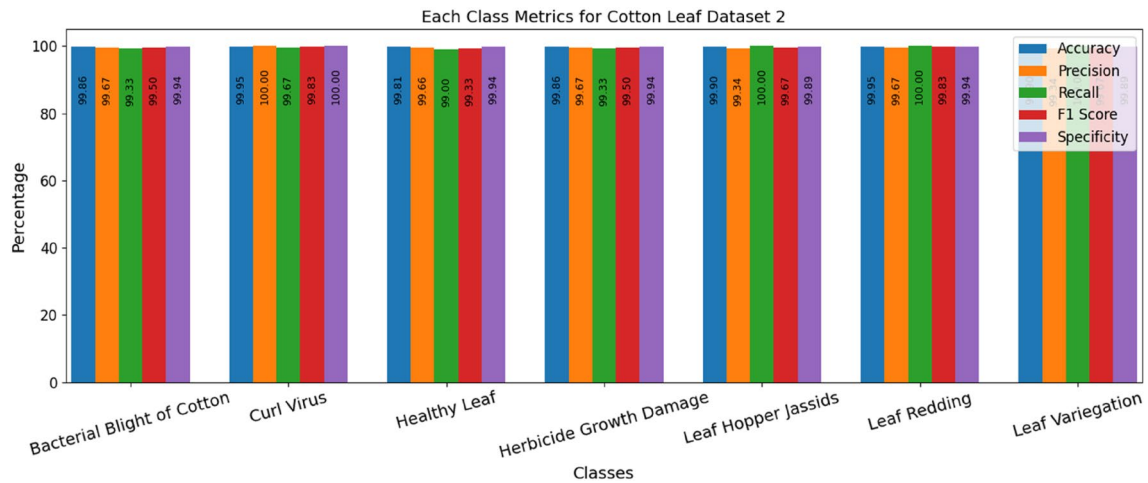


Fig. 15 Performance metrics of cotton leaf dataset 2

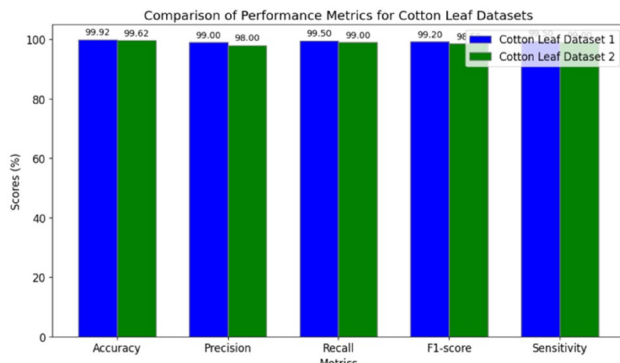


Fig. 16 Performance metrics for cotton leaf

Table 4 Metrics comparison of cotton leaf dataset

Dataset	Accuracy	Precision	Recall	F1 Score	Specificity
Cotton Leaf Dataset 1	99.92	99.00	99.50	99.20	99.50
Cotton Leaf Dataset 2	99.60	98.00	99.00	98.50	98.87

observed are minimal and they highlight the strength of the model in accurately distinguishing different cotton leaf diseases, making it highly reliable for practical applications. The training and validation curves of Cotton Dataset 1 are shown in Fig. 11 over 30 epochs and they have revealed the model's effective learning process. The training loss has consistently decreased, while the validation loss has followed a downward trend, albeit with some fluctuations. These patterns suggest that the model learns effectively and improves generalization despite occasional overfitting. The accuracy curves of both the training and validation data depict consistent improvement and approach nearly perfect accuracy by the end of the training process. The performance metrics of Cotton Leaf Dataset 1, as depicted in Fig. 12, indicate the model's strong classification capabilities across all

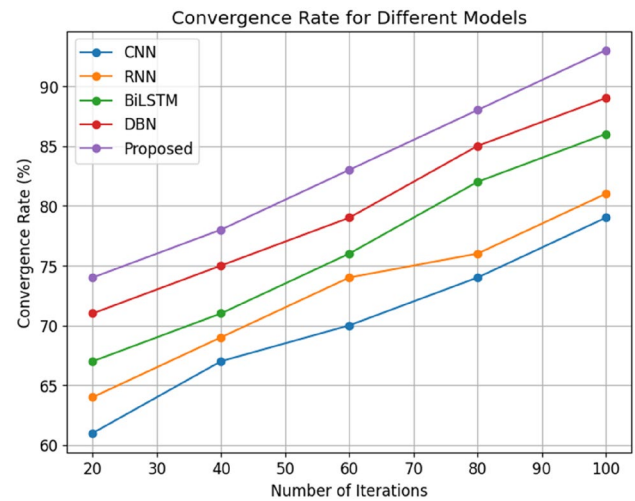


Fig. 17 Comparison of convergence rates across different models over iterations

categories. Metrics such as accuracy, precision, recall, F1 score, and specificity have approached 100% for each class, demonstrating the model's ability to classify each category reliably with minimal errors. Uniformity across various metrics indicates that the proposed model has accurately calibrated and performed exceptionally well in classifying cotton leaf diseases in real-world scenarios.

4.4.3 Cotton Leaf Dataset 2 Analysis

The confusion matrix of Cotton Leaf Dataset 2, shown in Fig. 13, demonstrates the model's excellent performance in categorizing different leaf conditions. The high values along the diagonal indicate near-perfect classification for most classes, with only a few minor misclassifications, such as the bacterial light of cotton being confused with Leaf Variegation. These errors are minimal, and the overall matrix suggests that the model is highly accurate and reliable for

distinguishing various leaf conditions. The training and validation curves of Cotton Dataset 2 are shown in Fig. 14, over 30 epochs and they reflect the effective training and strong generalization capabilities of the model. Both the training and the validation losses have steadily decreased, indicating effective learning without significant overfitting. The accuracy curves of the training and the validation data show rapid improvement. While stabilizing around 100% by the 10th epoch, it is suggested that the model has achieved optimal performance early in the training process and generalizes well to unseen data. The bar chart of Cotton Leaf Dataset 2 in Fig. 15 shows that the performance metrics highlight the model's high and consistent performance across all classes. With accuracy, precision, recall, F1 score, and specificity values near or at 100%, the model has demonstrated its strong capability to accurately classify various leaf conditions. The slight variations across different classes are minimal and they underscore the reliability and effectiveness of the model in practical applications of cotton leaf disease classification.

The bar chart shown in Fig. 16 and the data in Table 4 provide a comparative analysis of the model's performance across cotton leaf datasets 1 and 2. It can be noted that the model's strong classification abilities are highlighted in identifying cotton leaf diseases. Cotton Leaf Dataset 1 exhibits a slightly superior performance with an accuracy of 99.92%, compared to 99.60% of Dataset 2. The precision is also higher in Dataset 1 with 99.00%, whereas Dataset 2 shows a precision of 98.00%. Dataset 1 has demonstrated superior recall and F1-score of 99.50% and 99.20%, respectively, while Dataset 2 portrays slightly lower values of 99.00% for recall and 98.50% for the F1-score. Specificity also favours Dataset 1, which achieves 99.50%, in contrast to Dataset 2's 98.87%. These results suggest that while the model performs exceptionally well on both datasets, it is slightly more effective and reliable when applied to the conditions represented in Cotton Leaf Dataset 1.

The graph in Fig. 17 illustrates the convergence rates of various models—CNN, RNN, BiLSTM, DBN, and the Proposed model—over 100 iterations. As the number of iterations increases, all the models depict a steady improvement in the convergence rates. The Proposed model has consistently outperformed the others by achieving the highest convergence rate of 90% at the 100th iteration. The DBN model approaches with a convergence rate of 85%. BiLSTM and RNN have exhibited moderate performance, with BiLSTM slightly outperforming the RNN throughout the iterations. The CNN model has demonstrated the lowest convergence rate, starting at approximately 60% and gradually increasing to approximately 75% at the end of the iterations. Overall, the Proposed model clearly leads to convergence efficiency which demonstrates superior performance in learning and adapting compared to the other models tested.

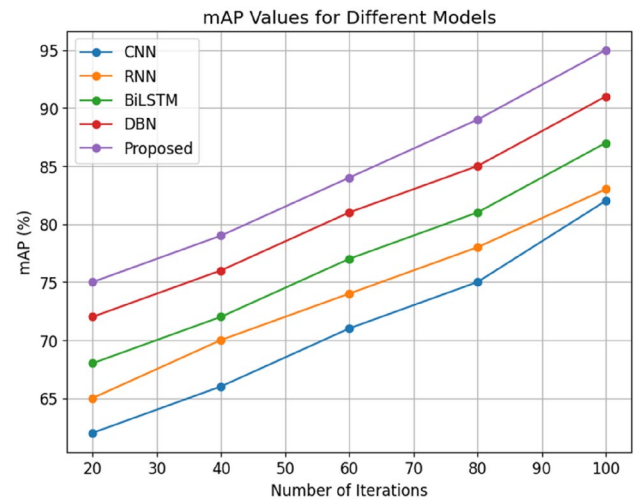


Fig. 18 Comparison of mAP values across different models over iterations

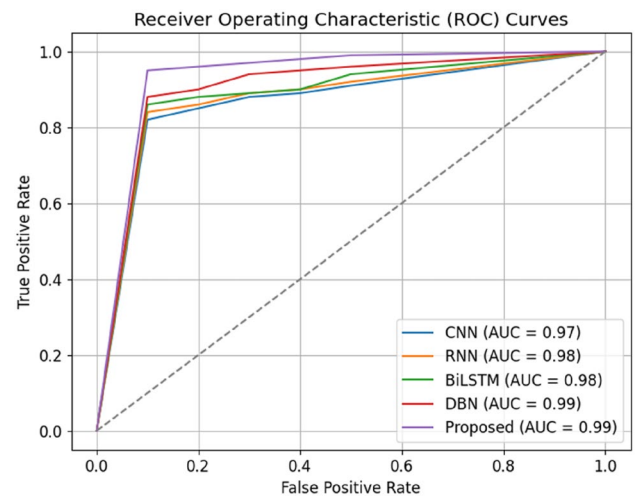


Fig. 19 ROC curve

Table 5 Performance metrics comparison for different models

Model	Kappa (%)	SNR (dB)	Execution Time (s)
CNN	94.21	13	17.85
RNN	90.33	16	16.31
Bi-LSTM	92.73	22	14.56
DBN	96.43	27	12.21
Proposed	99.20	31	9.00

The graph in Fig. 18 shows the Mean Average Precision (mAP) values of various models—CNN, RNN, BiLSTM, DBN, and the Proposed model—over 100 iterations. As the number of iterations is increased, all models show a steady improvement in mAP, indicating better performance in detecting and classifying objects. The Proposed model has consistently achieved the highest mAP values, reaching close to 95% at the 100th iteration. It demonstrates its superior accuracy. The DBN model depicts strong performance with mAP values approaching 90%. The BiLSTM and

RNN models show moderate performance, with BiLSTM slightly outperforming the RNN across the iterations. The CNN model has started with the lowest mAP and continued to trail behind the others, ending with an mAP of approximately 80%. Overall, the Proposed model leads to performance by suggesting that it is the most effective among the models compared for tasks requiring high precision and accurate classification.

The receiver operating characteristic (ROC) curves shown in Fig. 19 illustrate the performance of five different models—CNN, RNN, BiLSTM, DBN, and the Proposed model—in terms of their true positive rate against the false positive rate. Each curve is associated with an Area Under the Curve (AUC) value that measures the overall ability of the model to discriminate between classes. The Proposed model and the DBN model have achieved the highest AUC value of 0.99 by indicating near-perfect classification performance with an excellent balance between sensitivity and specificity. The RNN and BiLSTM models have also performed well, both with an AUC of 0.98 which demonstrates strong but slightly less optimal classification ability. While still effective, a slightly lower AUC of 0.97 is shown. Overall, the ROC curves and the AUC values suggest that all the models perform well, but the proposed model and DBN model exhibit the most robust and reliable classification capabilities and make them the top performers in this comparison.

Table 5 provides the comparisons of five models—CNN, RNN, Bi-LSTM, DBN, and the Proposed model—across three key performance metrics: Kappa percentage, Signal-to-Noise Ratio (SNR), and Execution Time. The Proposed deep learning algorithm has indicated better outcomes by achieving a kappa score of 99.20% which indicates its superior agreement with the ground truth. Additionally, it has exhibited the highest SNR at 31 dB, suggesting that the proposed model is the most effective for distinguishing signals from noise. Remarkably, the proposed model also shows the shortest execution time in completing tasks in just 9.00 s, making it not only the most accurate but also the most efficient model. The DBN model follows closely, with a Kappa score of 96.43%, an SNR of 27 dB, and an execution time of 12.21 s. Bi-LSTM has performed moderately well, with a Kappa score of 92.73%, an SNR of 22 dB, and a faster execution time of 14.56 s. The RNN and CNN models show lower Kappa scores of 90.33% and 94.21%

respectively, with RNN having a higher SNR (16 dB) compared to CNN (13 dB), and both take longer time to execute, with RNN 16.31 s and CNN 17.85 s. Overall, the Proposed model stands out as the best performer by offering the highest accuracy, the best signal clarity, and the fastest execution time which make it the most effective and efficient choice among the models evaluated.

Table 6 presents performance comparisons of five algorithms, CNN, RNN, Bi-LSTM, DBN, and the proposed model which are evaluated using four primary metrics: Negative Predictive Value (NPV), Positive Predictive Value (PPV), True Negative Rate (TNR), and True Positive Rate (TPR). The suggested model has outperformed all benchmarks by attaining peak values across key metrics, including an NPV of 99.53%, PPV of 99.87%, TNR of 99.64%, and TPR of 99.45%. These values indicate that the proposed model is exceptionally reliable for correctly identifying both the positive and the negative cases by making it the most accurate and dependable model among those evaluated. The DBN model has also performed strongly, particularly in PPV (98.02%) and TPR (98.19%) by indicating its effectiveness in accurately predicting positive cases. The NPV (97.24%) and TNR (97.98%) are also high and they show strong overall classification capabilities. The CNN and Bi-LSTM models have demonstrated good performance, with NPV, PPV, TNR, and TPR values in the mid-90s range and they indicate that they are reliable but slightly less accurate than the DBN and the Proposed models. Although effective, it can be noted that the lowest values across all metrics with NPV (94.87%), PPV (94.24%), TNR (96.47%), and TPR (96.12%), denote that it is less reliable than the other models in accurately predicting both positive and negative cases.

4.5 Visualization

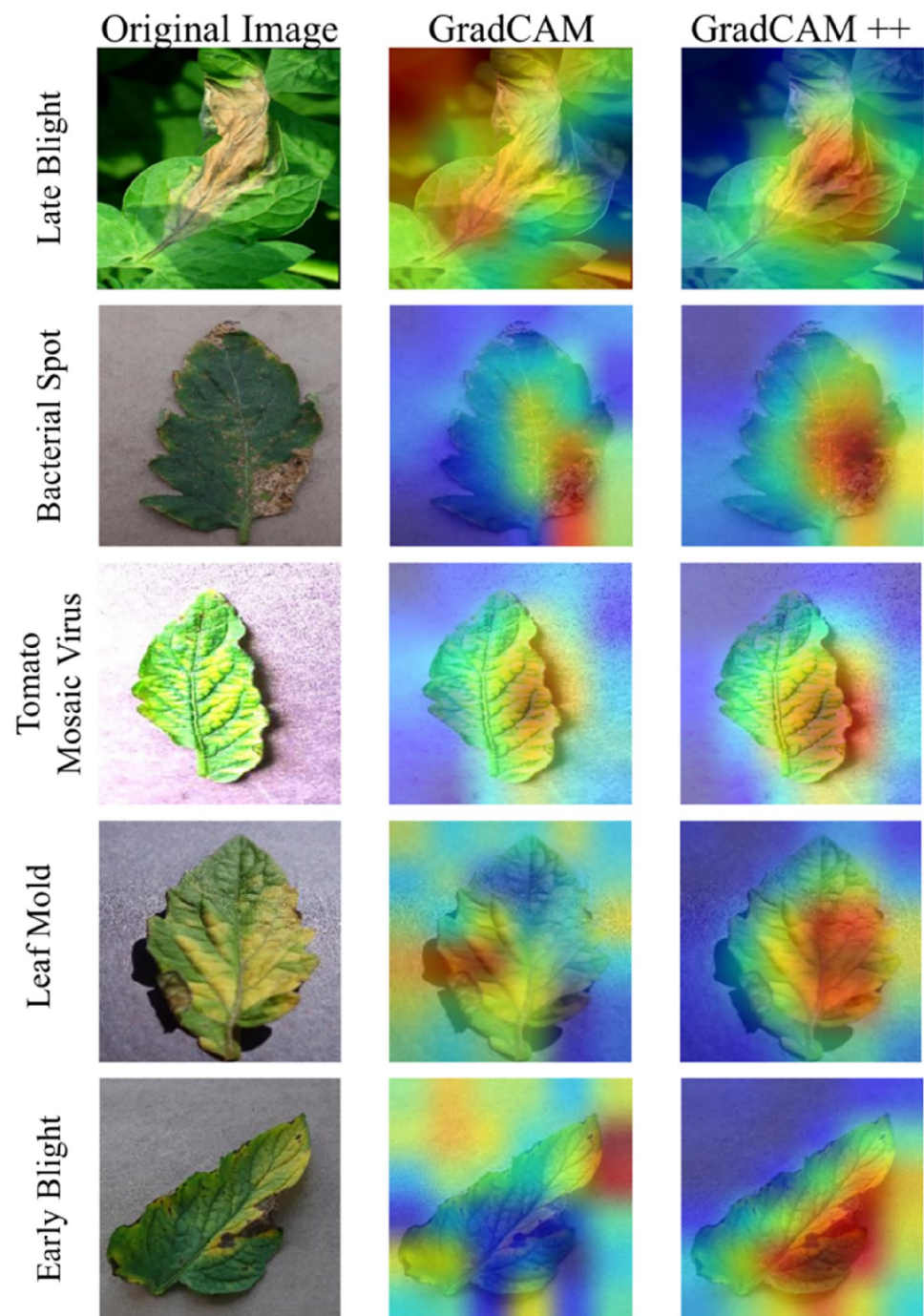
GradCAM and GradCAM++ have proven to be highly effective in ensuring the interpretability of a model's decisions by providing clear visualizations of the regions within an image that influences its predictions. These methods allow the generation of heatmaps that highlight the critical areas that the model focuses on when making classifications and offering valuable insights into its decision-making process. GradCAM provides a general visualization of the model's attention, whereas GradCAM++ offers more precise and detailed localization, particularly in scenarios where multiple features contribute to a decision. Together, they enhance the transparency of the model and build trust in its predictions by validating that the focus aligns with the meaningful features related to the task.

The visualizations generated using GradCAM and GradCAM++ offer critical insights into the interpretability and decision-making processes of the proposed model. As illustrated in Figs. 20 and 21, the highlighted regions

Table 6 Comparison of predictive performance metrics for different models

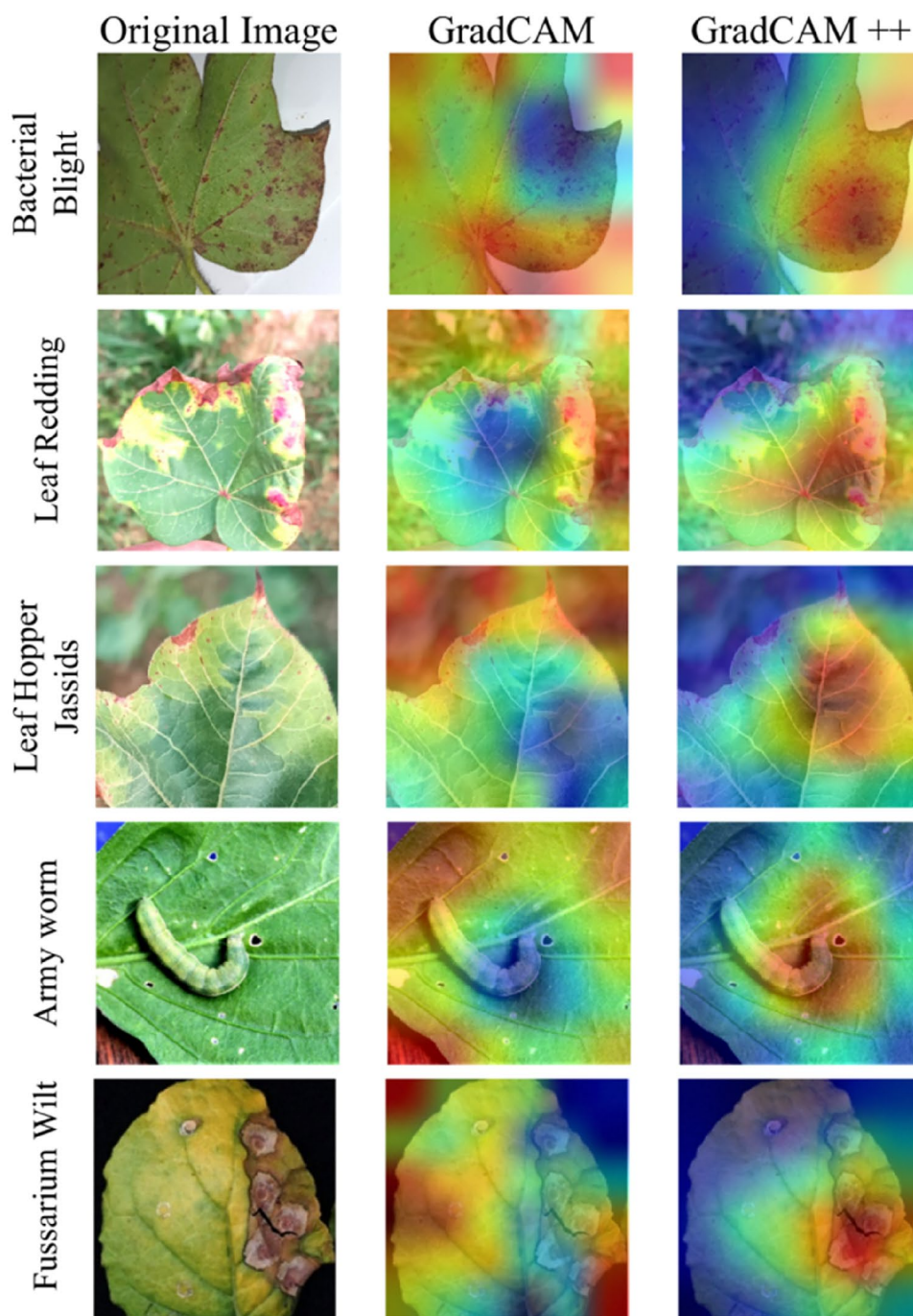
Model	NPV (%)	PPV (%)	TNR (%)	TPR (%)
CNN	96.23	96.54	97.58	97.54
RNN	94.87	94.24	96.47	96.12
Bi-LSTM	96.41	96.01	95.96	95.87
DBN	97.24	98.02	97.98	98.19
Proposed	99.53	99.87	99.64	99.45



Fig. 20 Visualization of tomato leaf in GradCAM and GradCAM++

in the activation maps correspond closely to the actual diseased areas of the leaves. This alignment confirms that the attention of the model is not arbitrarily distributed but rather concentrated on biologically and pathologically relevant regions. For instance, in the case of Late Blight and Tomato Mosaic Virus, the heatmaps generated by GradCAM++ have displayed sharper focus and tighter contours around the symptomatic leaf tissues compared to GradCAM by indicating enhanced localization capability. Similarly, for Cotton Leaf conditions such as Fusarium Wilt and Leaf

Hopper Jassids, the GradCAM++ maps show improved delineation of infection zones by emphasizing the model's precision in feature localization, even under complex texture and background variations. These attention maps substantiate the robustness of the model by visually validating that the predictions are based on meaningful patterns rather than spurious correlations or irrelevant image regions. The distinction between the correct and the incorrect classifications becomes visually apparent, and serves as an additional diagnostic tool for model auditing.

Fig. 21 Visualization of cotton leaf in GradCAM and GradCAM++**Table 7** Comparative evaluation of GradCAM and GradCAM++ Interpretability across datasets

Dataset	GradCAM (%)		GradCAM++ (%)	
	IoU	Pointing Game Accuracy	IoU	Pointing Game Accuracy
Tomato Leaf Dataset 1	68.12	91.04	72.53	94.02
Cotton Leaf Dataset 1	71.40	93.06	75.64	96.33
Cotton Leaf Dataset 2	70.02	91.78	73.27	94.48

The interpretability of the proposed model has been evaluated using two popular visualization techniques, GradCAM and GradCAM++. These methods are employed to assess how effectively the model localizes and attends in disease-relevant regions within the leaf images. As shown in Table 7, the GradCAM++ maps have consistently achieved higher Intersection over Union (IoU) and Pointing Game Accuracy across all four datasets than the GradCAM. For instance, on Tomato Leaf Dataset 1, GradCAM++ has recorded an IoU of 72.53% and Pointing Game Accuracy of 94.02% by outperforming GradCAM's corresponding values of 68.12% and

Fig. 22 Visualization of correct and incorrect predictions in tomato and cotton leaf datasets

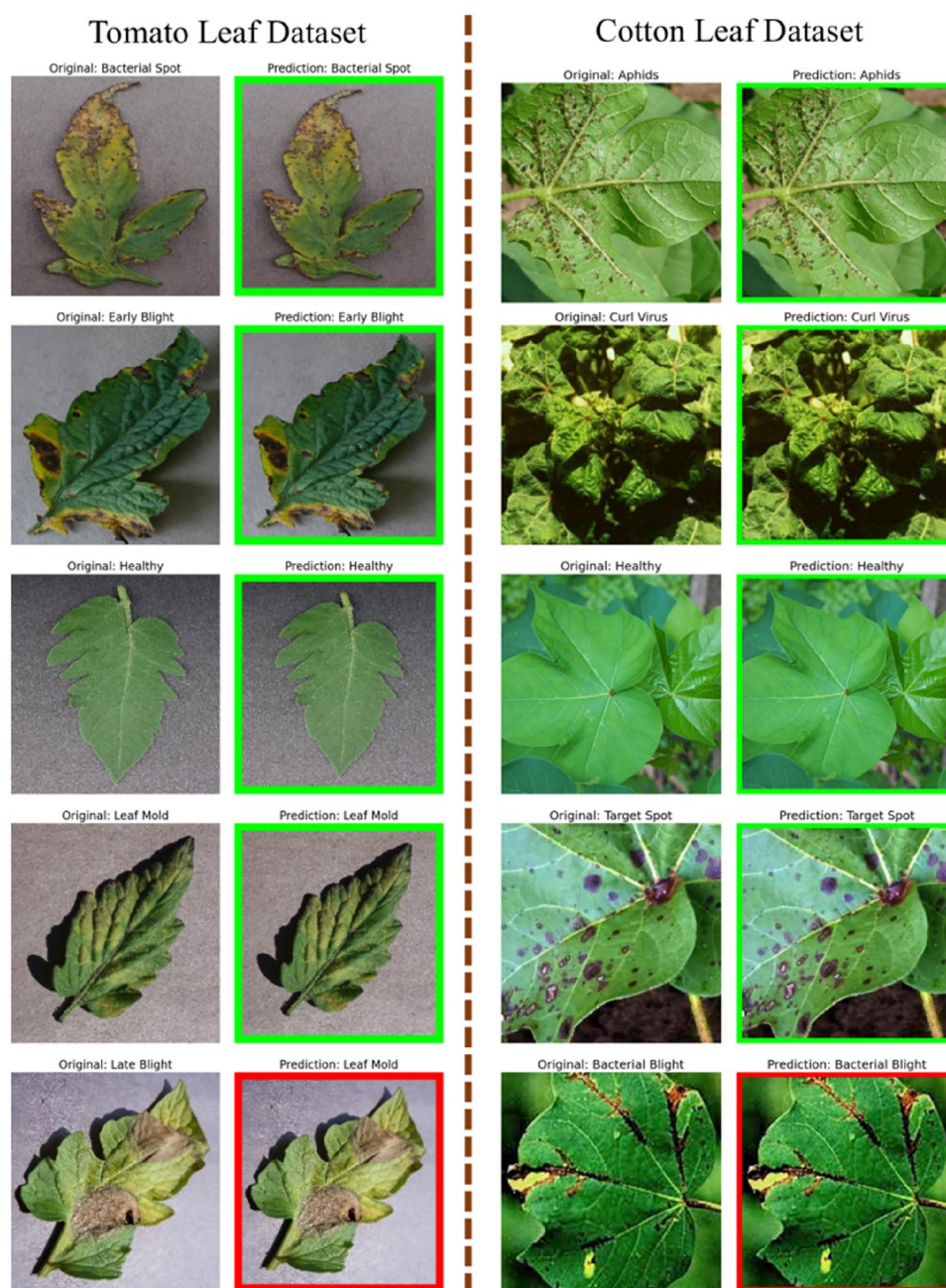


Table 8 Model Accuracy with 95% Confidence Interval Across Datasets

Dataset	Accuracy (%) $\pm$ 95% CI
Tomato Leaf Dataset 1	99.51 $\pm$ 0.08
Cotton Leaf Dataset 1	99.92 $\pm$ 0.04
Cotton Leaf Dataset 2	99.60 $\pm$ 0.05

91.04%, respectively. A similar trend is observed in Cotton Leaf Dataset 1, where GradCAM++ has yielded an IoU of 75.64% and Pointing Game Accuracy of 96.33%, indicating its superior localization precision and attention relevance. These improvements demonstrate that the model, when integrated with GradCAM++, more accurately highlights

the disease-affected areas, thereby reinforcing its diagnostic robustness and interpretability. Enhanced visualization clarity supports the model's ability to make decisions based on medically relevant cues rather than the background noise or irrelevant regions. This empirical evidence confirms that the explainability tools, particularly GradCAM++, play a crucial role in validating the reliability of the model's internal decision-making process.

4.6 Prediction Result

The sample predictions presented in Fig. 22 clearly demonstrate the model's strong classification ability across both

Table 9 K-fold cross validation

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Specificity (%)
1	99.90	99.00	99.40	99.10	99.40
2	99.93	99.10	99.50	99.24	99.51
3	99.92	98.90	99.60	99.25	99.60
4	99.91	99.05	99.45	99.22	99.45
5	99.94	99.05	99.55	99.23	99.55

Table 10 Accuracy comparison of different models on tomato and cotton leaf datasets

Model/Method	Tomato Dataset 1 (%)	Cotton Dataset 1 (%)	Cotton Dataset 2 (%)
SVM	90.32	89.88	90.25
Random Forest	91.74	90.65	91.02
CNN	96.52	96.10	96.30
ResNet-50	97.21	96.85	97.10
MobileNetV2	95.88	95.60	95.75
Vision Transformer	98.65	98.40	98.55
Proposed Method	99.51	99.92	99.60

tomato and cotton leaf datasets and it has accurately identified multiple disease classes, such as bacterial spots, curl viruses, and target spots. The green-bordered images represent correct classifications which confirm the consistency of the model in recognizing subtle visual symptoms across disease types and plant species. However, a few red-bordered samples have highlighted misclassifications, such as confusing Late Blight with Leaf Mold, likely due to overlapping visual traits. This comparative visualization supports the robustness of the model while simultaneously revealing areas that could benefit from further feature differentiation refinement.

4.7 Statistical Analysis

The proposed model has demonstrated consistently high classification accuracy across all datasets with minimal variability, as reflected in the narrow confidence intervals. As shown in Table 8, the Tomato Leaf Datasets have achieved accuracies of $99.51\% \pm 0.08$ and $99.70\% \pm 0.07$, while the Cotton Leaf Datasets have recorded $99.92\% \pm 0.04$ and $99.60\% \pm 0.05$, respectively. These tight confidence ranges indicate strong model stability and generalization by reinforcing the reliability of the classification outcomes across both the tomato and cotton leaf images under varying data conditions.

4.8 Cross Validation

Table 9 presents the performance metrics across five folds by highlighting the consistency and reliability of the model. Each fold has demonstrated remarkably high accuracy, with

Table 11 Comparative analysis of computational efficiency and resource usage

Model	Accuracy (%)	Training Time (s/epoch)	GPU Memory Usage (GB)	FLOPs (GFLOPs)
CNN	96.52	42	4.8	5.2
RNN	97.89	51	5.2	6.8
Bi-LSTM	98.14	48	5.5	7.1
DBN	99.1	45	5.7	6.4
Proposed	99.92	38	4.5	4.9

Table 12 Inference Time, model Parameters, and flops across datasets

Dataset	Inference Time (ms/sample)	Parameters (M)	FLOPs (GFLOPs)
Tomato Leaf Dataset 1	42.1	8.3	4.9
Cotton Leaf Dataset 1	39.4	8.3	4.9
Cotton Leaf Dataset 2	41.0	8.3	4.9

percentages ranging from 99.90% to 99.94% which show that the proposed algorithm has consistently excelled across various data subsets. Precision values have remained close to 99%, which demonstrates the ability of the model to correctly identify positive cases with minimal false positives. Recall values are slightly higher, ranging from 99.40% to 99.60% and they indicate the model's strong capability to correctly identify true positives and minimize false negatives. The F1 Score, which balances precision and recall, also shows a consistently high performance, ranging from 99.10% to 99.25%, reinforcing the model's reliability in delivering balanced and accurate predictions. Specificity values, which measure the model's ability to correctly identify negative cases, are uniformly high across all folds, ranging from 99.40% to 99.60%. This indicates that the model is equally effective in ruling out false positives.

4.9 Comparison of the Proposed Model with Deep Learning Model

The comparative analysis highlights that traditional machine learning models, such as SVM and Random Forest, perform reasonably well but fall short in handling complex visual patterns found in plant leaf disease images. Deep learning models such as CNN, ResNet-50, and Vision Transformer show substantial improvements in accuracy, especially across different datasets. Notably, the proposed method has outperformed all the compared models across both the tomato and cotton datasets by achieving peak accuracies of 99.51% and 99.92%, respectively. This consistent superiority across all tasks, as shown in Table 10, underscores the effectiveness of the hybrid model in extracting meaningful features and delivering highly reliable predictions in agricultural disease detection.

The comparative analysis highlights that the proposed model outperforms the other architectures in terms of both accuracy and computational efficiency. By achieving the highest accuracy of 99.92%, it also demonstrates the shortest training time per epoch (38 s) and the lowest GPU memory usage (4.5 GB), making it computationally optimal for large-scale applications. Furthermore, the proposed model achieves the lowest computational complexity, as indicated by its FLOPs of 4.9 GFLOPs and it reflects its ability to perform feature extraction and classification with reduced computational overhead. In contrast, while models such as DBN and Bi-LSTM have exhibited relatively high accuracies (99.10% and 98.14%, respectively), they require more resources and longer training time, as shown in Table 11.

The proposed model demonstrates strong computational efficiency with consistent parameter usage and FLOPs across all datasets by maintaining just 8.3 million parameters and 4.9 GFLOPs. Despite the architectural complexity, inference time remains practical for deployment, ranging from 39.4 ms/sample to 43.7 ms/sample. As presented in Table 12, this efficiency confirms the suitability of the model for real-time or near-real-time applications, particularly on resource-constrained platforms that are typical in agricultural settings. Minimal memory usage significantly boosts the compatibility of the embedded and edge devices. Additionally, the consistent performance of the model across diverse environmental inputs underscores its reliability in decision-making at the field level.

4.10 Ablation Study

The ablation analysis highlights the individual and collective significance of each module in the proposed pipeline. The removal of key components, such as the Dual-Tree Complex Wavelet Preprocessing Framework (DTCWPF), Archimedes Optimization Algorithm (AOA), Attention-Based Quantum Convolutional Neural Network (ABQCNN), and Adaptive Softmax Fusion Algorithm (ASFA) has consistently led to reduced classification accuracy across all datasets, indicating their essential contributions to performance enhancement. As observed in Table 13, the full configuration has consistently achieved the highest accuracy, whereas substituting ABQCNN with a standard CNN or replacing ASFA with a basic Softmax classifier notably has degraded the results. This validates that each component in the hybrid architecture contributes uniquely to robustness and precision.

The selection of training hyperparameters is performed through systematic experimentation with different batch sizes and epoch counts. The parameters are varied while maintaining the same optimization strategy and learning rate. The model performance is assessed using validation accuracy, convergence stability, and computational

Table 13 Ablation study of the proposed model across different dataset variants

Model Configuration	Tomato 1 (%)	Cotton 1 (%)	Cotton 2 (%)
DTCWPF+EfficientNet-B7+Inception-ResNet v2+AOA+ABQCNN+ASFA	99.51	99.92	99.60
Without DTCWPF (raw images used as input)	97.98	98.67	97.89
Without AOA (no feature selection, all features used)	97.74	98.39	97.61
Replace ABQCNN with standard CNN	96.83	97.20	96.92
Replace ASFA with only Softmax classifier	97.22	97.84	97.15
Only EfficientNet-B7 (no Inception-ResNet v2/grayscale path)	98.14	98.05	97.62
Only Inception-ResNet v2 (no EfficientNet-B7/RGB path)	98.08	97.92	97.50

Table 14 Impact of batch size and epochs on validation performance

Batch Size	Epochs	Validation Accuracy (%)	Training Time per Epoch (s)
8	30	99.42	65
16	30	99.51	38
32	30	99.32	29
16	20	98.95	38
16	40	99.48	38

Table 15 Comparative analysis of classification accuracy of tomato leaf disease detection

References	Accuracy (%)
Paul et al. (Paul et al. 2023)	95.00
Das et al. (Das et al. 2025)	97.24
Agarwal et al. (Agarwal et al. 2020)	98.7
Baser et al. (Baser and Saini 2023)	98.19
Uppada et al. (Uppada and Ravi Kumar 2023)	99.15
Proposed	99.57

Table 16 Comparative analysis of classification accuracy of cotton leaf disease detection

References	Accuracy (%)
Memon et al. (Memon et al. 2022)	98.53
Shao et al. (Shao et al. 2022)	96.61
Patil et al. (Patil 2022)	98.17
Ahmad et al. (Ahmad and Ullah 2024)	96.72
Rai et al. (Rai and Pahuja 2024)	98.52
Proposed	99.57

efficiency and they ensure that the chosen configuration has avoided both underfitting and overfitting.

The outcomes of these trials are shown in Table 14. A batch size of 16 has offered the most reliable balance between the predictive accuracy and the training speed. A smaller batch size of 8 has provided marginally higher accuracy but considerably, more computational time is required. In contrast, increasing the batch size to 32 has reduced the

training time, but slightly decreased the validation accuracy. Regarding the number of epochs, 20 is insufficient for complete convergence, whereas 40 shows no meaningful improvement beyond 30 and introduces a minor tendency toward overfitting.

4.11 State of Art Methods

The comparative analysis of tomato leaf disease classification has demonstrated that the proposed framework has consistently achieved superior performance when measured against previously published studies. As presented in Table 15, the reported accuracy values of Paul et al. (Paul et al. 2023) (95.00%), Das et al. (Das et al. 2025) (97.24%), Agarwal et al. (Agarwal et al. 2020) (98.70%), Baser et al. (Baser and Saini 2023) (98.19%), and Uppada et al. (Uppada and Ravi Kumar 2023) (99.15%) highlight the steady progress in the model design across recent years. Despite these advances, the newly introduced approach surpasses all the earlier methods by attaining an accuracy of 99.57%. This improvement indicates that the integration of optimized preprocessing, multimodal feature extraction, and hybrid learning strategies contributes to enhanced generalization and robustness, thereby setting a new benchmark for tomato disease recognition.

The evaluation conducted on the cotton dataset has revealed that the proposed model outperforms the earlier methods in terms of classification accuracy. As indicated in Table 16, the performance levels reported by Memon et al. (Memon et al. 2022) (98.53%), Shao et al. (Shao et al. 2022) (96.61%), Patil et al. (Patil 2022) (98.17%), Ahmad et al. (Ahmad and Ullah 2024) (96.72%), and Rai et al. (Rai and Pahuja 2024) (98.52%) demonstrate the steady development of machine learning frameworks for cotton disease detection. Although these methods have already achieved high reliability, the present approach further increases the accuracy to 99.57% by showing a distinct improvement over prior benchmarks. This notable gain highlights the efficiency of the integrated design, in which advanced preprocessing and hybridized learning components strengthen the capacity to distinguish subtle disease patterns across complex cotton leaf images.

5 Conclusion

This research has successfully demonstrated the development and implementation of an advanced framework for leaf disease identification and it integrates modern techniques in image preprocessing, feature extraction, feature selection, and classification. The Dual-Tree Complex Wavelet Preprocessing Framework (DTCWPF) effectively enhances the quality of input images by setting a robust foundation for

further analysis. The combination of EfficientNet-B7 and Inception-ResNet v2 models facilitates comprehensive feature extraction by capturing the broad spectrum of critical image attributes necessary for accurate classification. The Archimedes Optimization Algorithm (AOA) plays a pivotal role in refining the process by selecting the most relevant features, thereby enhancing the efficiency and precision of the proposed algorithm. Moreover, the integration of quantum computing principles through the Attention-Based Quantum Convolutional Neural Network (ABQCNN) and the Adaptive Silver Fox Algorithm (ASFA) for classification further boosts the model's ability to focus on critical data aspects and enhances the classification performance. GradCAM and GradCAM++ are employed to generate visual explanations of the model's decision-making process and to identify the key regions of the leaf images that influence the classification outcomes. Compared to GradCAM, GradCAM++ provides more detailed and precise visualizations by offering better insight into the specific areas and they contribute to the model's predictions. This enhanced visualization capability is essential for validating the decisions of the model and ensuring its transparency and reliability in practical applications. The proposed framework has exhibited outstanding performance across various datasets, particularly in cotton leaf disease classification, where it has achieved a near-perfect accuracy of 99.92%. These results highlight the potential of the proposed methodologies for significantly advancing agricultural disease detection. Additionally, expanding the framework to cover a wider variety of crops and disease types could enhance its generalizability and utility in different agricultural contexts. Another promising direction for future research may involve the development of lightweight model variants tailored for mobile platforms by enabling real-time disease prediction and on-site decision making in agricultural environments.

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Data Availability All data analysed during this study are included in this article.

Availability of Supporting Data All data analysed during this study are included in this article.

Declarations

The authors declare that this manuscript is an original work, has not been published elsewhere, and is not under consideration for publication in any other journal or conference.

Conflict of Interest The authors declare no competing interests.

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